

Cohomological properness of sheaves of sets and of sheaves of non-commutative groups

SGA 1, Exposé XIII

English draft

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About this draft

This is an unofficial English translation of Exposé XIII, “Propreté cohomologique des faisceaux d’ensembles et des faisceaux de groupes non commutatifs”, from SGA 1 (after unpublished notes of A. Grothendieck). The source volume is A. Grothendieck and M. Raynaud, *Revêtements étales et groupe fondamental* (SGA 1), Séminaire de géométrie algébrique du Bois Marie, 1960–61. It follows the slightly corrected SMF recomposition, arXiv:math/0206203v2.

This draft contains the whole exposé, including proofs, footnotes, diagrams, the updating remarks added in 2003 by M. Raynaud (between bold brackets and marked (MR)) where the exposé has them, and its bibliography. Statement numbering and mathematical notation follow the source. References to other exposés print the source’s numbers. Apparent misprints in the source have been retained; they are recorded in the accompanying README, separately from the translated text. Scholarly proofreading remains outstanding.

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EXPOSÉ XIII

Cohomological properness of sheaves of sets and of sheaves of non-commutative groups

by Mme M. RAYNAUD¹

This exposé proposes to use étale cohomology in order to generalize certain results of IX and X. It also shows how one can extend to sheaves of not necessarily commutative groups those results of SGA 5 II which still make sense for such sheaves. The notions of étale cohomology expounded in SGA 4 are assumed known.

The main result (2.4) gives an important example of a non-proper morphism $f: U \rightarrow S$ which is “cohomologically proper in dimension ≤ 1 ”, that is to say such that, for certain sheaves of groups F on U (in the sense of the étale topology), the formation of f_*F and R^1f_*F commutes with every change of base $S' \rightarrow S$. This property is indeed satisfied by the open set U of a scheme X proper over S , complementary to a divisor D with normal crossings relative to S , at least if one requires F to be constant finite, of order prime to the residue characteristics of S . If one no longer assumes F of order prime to the residue characteristics of S , one has an analogous result on replacing R^1f_*F by the subsheaf $R_t^1f_*F$ obtained by restricting oneself to considering the torsors under F “tamely ramified on X relative to S ”. In particular this makes it possible to show that the tamely ramified fundamental group of a proper and smooth algebraic curve over a separably closed field, with a finite number of closed points removed, is topologically of finite type (2.12).

No. 4 is devoted to the homotopy exact sequence and to the Künneth formula.

Finally an appendix gives useful variants of Abhyankar’s lemma proved in X.3.6.

0. Recollections on the theory of stacks

We shall use in what follows the theory of stacks expounded in [1] and [2]. We restrict ourselves to the case of the étale site of a scheme. Given a scheme X , let us denote by $X_{\text{ét}}$ the étale site of X . Recall that a stack F on X is a fibered category over $X_{\text{ét}}$ such that, for every scheme X' étale over X and for every pair of objects x, y of the fiber $F_{X'}$, the presheaf $\mathbf{Hom}_{X'}(x, y)$ is a sheaf, and such that, for every surjective étale morphism $X'' \rightarrow X'$, every object of $F_{X''}$ endowed with a descent datum relative to $X'' \rightarrow X'$ is the inverse image of an object of $F_{X'}$.

We denote by $F(X')$ the category of cartesian sections of F/X' . More generally, if $\mathbf{Sch}_X$ is the category of schemes over X endowed with the étale topology, the stack F can be extended to a stack $\mathcal{F}$ on $\mathbf{Sch}_X$ and, for every morphism $f: X' \rightarrow X$, we again denote by $F(X')$ the category of cartesian sections of this stack $\mathcal{F}$ over X' .

A gerbe is a stack such that, for every scheme X' étale over X and for every pair of objects x, y of $F_{X'}$, every morphism from x to y is an isomorphism and x and y are locally isomorphic, and such that the set of objects X' of $X_{\text{ét}}$ such that $F_{X'}$ is nonempty is a refinement of $X_{\text{ét}}$. For example the stack of torsors under a sheaf of groups is a gerbe which, moreover, has a

¹After unpublished notes of A. Grothendieck.

cartesian section. Conversely a gerbe which has a section, i.e. such that there exists an object x of F_X , is equivalent to the stack of torsors under the sheaf of groups $\mathbf{Aut}_X(x)$.

There is an obvious notion of subgerbe and of maximal subgerbe of a stack F . Given a cartesian section x of $F(X)$, there exists a unique maximal subgerbe G_x of F such that x factors through G_x . We call G_x the subgerbe generated by x ; it is by definition a trivial gerbe. The presheaf SF defined by

$$SF(X') = \{\text{maximal subgerbes of } F|X'\}$$

is a sheaf, called the sheaf of maximal subgerbes of F . Let O be the presheaf defined by

$$O(X') = \{\text{classes of objects of } F_{X'} \text{ mod. isomorphism}\}.$$

By associating to every object x of $F_{X'}$ the maximal subgerbe of $F|X'$ generated by x , we obtain a morphism

$$O \longrightarrow SF;$$

by [2, III 2.1.4], this morphism makes SF a sheaf associated to O .

A stack F is said to be *constructible* (resp. *ind- $\mathbb{L}$ -finite*, $\mathbb{L}$ being a set of prime numbers) if, for every scheme X' étale over X and for every object x of $F_{X'}$, the same holds for the sheaf $\mathbf{Aut}_{X'}(x)$ [2, VII 2.2.1]. A stack is said to be 1-constructible if it is constructible and if the sheaf of maximal subgerbes is constructible.

1. Cohomological properness

1.0. Let S be a scheme, $f: X \rightarrow Y$ a morphism of S -schemes. If S' is an S -scheme, we consider the following diagram, all of whose squares are cartesian:

$$(1.0.1) \quad \begin{array}{ccc} X & \xleftarrow{h} & X' \\ f \downarrow & & \downarrow f' \\ Y & \xleftarrow{g} & Y' \\ \downarrow & & \downarrow \\ S & \xleftarrow{\quad} & S'. \end{array}$$

If Y_1 is a scheme étale over Y , we set $X_1 = X \times_Y Y_1$, $Y'_1 = Y' \times_Y Y_1$, and we consider the cartesian square

$$(1.0.2) \quad \begin{array}{ccc} X_1 & \xleftarrow{h_1} & X'_1 \\ f_1 \downarrow & & \downarrow f'_1 \\ Y_1 & \xleftarrow{g_1} & Y'_1. \end{array}$$

DEFINITION 1.1. Let F be a stack on X . We say that (F, f) is *cohomologically proper relative to S in dimension ≤ -1* (resp. in dimension ≤ 0 , resp. in dimension ≤ 1) if, for every S -scheme S' , the canonical functor (defined in the obvious way by the universal property of the inverse image of stacks):

$$g^* f_* F \longrightarrow f'_* h^* F \quad (\text{cf. 1.0.1})$$

is faithful (resp. fully faithful, resp. an equivalence of categories).

If there is no possible confusion about S , in particular if $S = Y$, we say cohomologically proper instead of cohomologically proper relative to S .

1.2. Let F be a sheaf of sets on X ; let Φ be the stack in discrete categories associated to F , i.e. the stack whose fiber over every scheme X_1 étale over X is the discrete category having $F(X_1)$ as its set of objects. We say that (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. in dimension ≤ 0) if (Φ, f) is cohomologically proper relative to S in dimension ≤ 0 (resp. in dimension ≤ 1).

The canonical morphism

$$(1.2.1) \quad g^* f_* F \longrightarrow f'_* h^* F$$

gives, on passing to the associated stacks in discrete categories, the canonical morphism

$$g^* f_* \Phi \longrightarrow f'_* h^* \Phi.$$

Consequently, to say that (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. in dimension ≤ 0) is equivalent to saying that, for every S -scheme S' , the morphism (1.2.1) is injective (resp. bijective).

1.3. Let F be a sheaf of groups on X and Φ the stack of torsors on X with group F [1, II 2.3.2]. We say that (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1) if (Φ, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1). The condition of cohomological properness can be made explicit as follows.

PROPOSITION 1.3.1. *The notation is that of (1.0.1) and (1.0.2). Let F be a sheaf of groups on X . We denote by F' (resp. F_1 , resp. F'_1 , etc.) the inverse image of F on X' (resp. on X_1 , resp. on X'_1 , etc.). Then the following conditions are equivalent:*

- (i) (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1).
- (ii) For every morphism $S' \rightarrow S$, for every scheme Y_1 étale over Y , and for every torsor P on X_1 with group F_1 , if ${}^P F_1$ denotes the group F_1 twisted by P [1, II 4.1.2.3], the canonical morphism

$$a_0: g_1^*(f_{1*}({}^P F_1)) \longrightarrow f'_{1*}({}^P F'_1)$$

is injective (resp. a_0 is bijective and the canonical morphism

$$a_1: g^*(R^1 f_* F) \longrightarrow R^1 f'_* F'$$

is injective, resp. a_0 and a_1 are bijective).

- (ii bis) For every morphism $S' \rightarrow S$, for every scheme Y_1 étale over Y , for every torsor P on X_1 with group F_1 , and for every torsor R under ${}^P F_1$, the canonical morphism

$$\alpha_0: g_1^*(f_{1*} R) \longrightarrow f'_{1*} R'$$

is injective (resp. α_0 is bijective, resp. the morphisms α_0 and

$$\alpha_1: g_1^*(R^1 f_{1*}({}^P F_1)) \longrightarrow R^1 f'_{1*}({}^P F'_1)$$

are bijective).

Proof. (i) $\Rightarrow$ (ii bis). By [1, II 4.2.5] every torsor R with group ${}^P F_1$ is of the form $R = Q \overset{F_1}{\wedge} P^\circ$, where Q is a torsor with group F_1 and P° the opposite of P . We then have $R' \simeq Q' \overset{F'_1}{\wedge} P'^\circ$. Let Φ be the stack of torsors under F and let x, y (resp. x', y') be the objects of the fiber category $(g^* f_* \Phi)_{Y'_1}$ (resp. $(f'_* \Phi')_{Y'_1}$) associated to P, Q (resp. P', Q'). We have the relation

$$Q \overset{F_1}{\wedge} P^\circ \simeq \mathbf{Hom}_{F_1}(P, Q),$$

and it follows that we have canonical isomorphisms

$$\mathbf{Hom}_{Y'_1}(x, y) \simeq g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ), \quad \mathbf{Hom}_{Y'_1}(x', y') \simeq f'_{1*}(Q' \overset{F'_1}{\wedge} P'^\circ).$$

Consequently the morphism α_0 is identified with the morphism

$$\mathbf{Hom}_{Y'_1}(x, y) \longrightarrow \mathbf{Hom}_{Y'_1}(x', y'),$$

from which it follows that, if (F, f) is cohomologically proper relative to S in dimension ≤ -1 , α_0 is injective and that, if (F, f) is cohomologically proper in dimension ≤ 0 , α_0 is bijective.

Suppose now that (F, f) is cohomologically proper relative to S in dimension ≤ 1 , i.e. that the canonical morphism

$$\varphi: g^*f_*\Phi \longrightarrow f'_*\Phi'$$

is an equivalence. Let G be the sheaf of maximal subgerbes of the stack $f_*\Phi$ [1, III 2.1.8]; we then have an isomorphism $G \simeq R^1f_*F$. Since g^*G is the sheaf of maximal subgerbes of $g^*f_*\Phi$ [2, III 2.1.5.5], the morphism α_1 is obtained from $\varphi|_{Y'_1}$ by taking the sheaves of maximal subgerbes, hence is an isomorphism.

(ii bis) $\Rightarrow$ (ii). It suffices to show that, if the morphisms α_0 are bijective, then the morphisms α_1 are injective. Let Y'_1 be a scheme étale over Y' , and s and t two elements of $g^*(R^1f_*F)(Y'_1)$ having the same image in $R^1f'_*F'(Y'_1)$, and let us show that we have $s = t$. The assertion is local for the étale topology of Y'_1 and, taking into account the definition of the inverse image $g^*(R^1f_*F)$, we may assume that Y'_1 is the inverse image of a scheme Y_1 étale over Y and that s and t come from torsors P and Q on X_1 . The hypothesis made on s and t then means that the inverse images P' and Q' of P and Q on X'_1 are isomorphic locally for the étale topology of Y'_1 . If we set $R = \mathbf{Hom}_{F_1}(P, Q)$, the fact that the morphism

$$g_1^*f_{1*}R \longrightarrow f'_{1*}R'$$

is bijective proves that P and Q are isomorphic locally for the étale topology of Y_1 , hence that we have $s = t$.

(ii) $\Rightarrow$ (i). To prove that φ is faithful (resp. fully faithful), it suffices to show that, if Y_1 is a scheme étale over Y , if P, Q are two torsors on X_1 with group F_1 , and if x, y (resp. x', y') are the objects of $(g^*f_*\Phi)_{Y'_1}$ (resp. $(f'_*\Phi')_{Y'_1}$) associated to P, Q (resp. P', Q'), then the morphism

$$a: \mathbf{Hom}(x, y) \longrightarrow \mathbf{Hom}(x', y')$$

is injective (resp. bijective). Now a is identified with the canonical morphism

$$H^0(Y'_1, g_1^*f_{1*}(Q \overset{F_1}{\wedge} P^\circ)) \longrightarrow H^0(Y'_1, f'_{1*}(Q' \overset{F'_1}{\wedge} P'^\circ)).$$

If we have $\mathbf{Hom}(x, y) \neq \emptyset$, then $Q \overset{F_1}{\wedge} P^\circ$ is a torsor under PF_1 which is locally trivial over Y_1 ; it follows that $f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a torsor under $f_{1*}({}^PF_1)$, and that $g_1^*f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a trivial torsor. The morphism a is then identified with the canonical morphism

$$H^0(Y_1, g_1^*f_{1*}({}^PF_1)) \longrightarrow H^0(Y'_1, f'_{1*}({}^{P'}F'_1)).$$

The same holds if we have $\mathbf{Hom}(x', y') \neq \emptyset$ and if a_1 is injective, for then $Q' \overset{F'_1}{\wedge} P'^\circ$ is trivial, and it follows from the injectivity of a_1 that P and Q are locally isomorphic over Y_1 . We conclude that, if a_0 is injective (resp. if a_0 is bijective and a_1 injective), φ is faithful (resp. fully faithful).

It remains to show that, if a_0 and a_1 are bijective, the functor φ is essentially surjective. Let Y'' be a scheme étale over Y' , $X'' = X' \times_{Y'} Y''$, and let P'' be a torsor on X'' with group $F'' = F'|_{X''}$. We shall show that there exists an element x of $(g^*f_*\Phi)_{Y''}$ whose image in $(f'_*\Phi')_{Y''}$ is isomorphic to P'' . Let p'' be the class of P'' . From the fact that a_1 is surjective it follows that one can find a surjective étale morphism $Y''_1 \rightarrow Y''$, an étale morphism $Y_1 \rightarrow Y$ such that there is a morphism $Y''_1 \rightarrow Y'_1$, and a torsor P_1 on X_1 with group F_1 whose inverse image P''_1 on X''_1 is isomorphic to the inverse image of P'' . Using the fact that φ is fully faithful, one sees that the object x_1 of $(g^*f_*\Phi)_{Y''_1}$ which corresponds to P''_1 is endowed with a descent datum relative to $Y''_1 \rightarrow Y''$, hence comes from an element x of $(g^*f_*\Phi)_{Y''}$. Since the

image of x in $(f'_*\Phi')_{Y''}$ is P'' , this proves that φ is essentially surjective and completes the proof.

EXAMPLE 1.4. Let $f: X \rightarrow Y$ be a proper morphism. It follows from [2, VII 2.2.2] that, for every ind-finite stack F on X , the pair (F, f) is cohomologically proper (relative to Y) in dimension ≤ 1 . In particular, for every sheaf of sets (resp. every sheaf of groups, resp. every ind-finite sheaf of groups) F on X , (F, f) is cohomologically proper in dimension ≤ 0 (resp. in dimension ≤ 0 , resp. in dimension ≤ 1).

REMARKS 1.5. a) Let F be a sheaf of groups on X such that (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0). If one regards F as a sheaf of sets, (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0), but the converse is false.

For example, let Y be the spectrum of a strictly local discrete valuation ring with closed point t and generic point s , $f: X \rightarrow Y$ a nonempty scheme over Y whose closed fiber is empty, F a nontrivial constant sheaf of groups on X , and P a torsor under F such that we have $H^0(X_s, {}^P F|_{X_s}) = 1$. Then $({}^P F, f)$ is cohomologically proper relative to Y in dimension ≤ -1 when one regards ${}^P F$ as a sheaf of sets. If one regards ${}^P F$ as a sheaf of groups, one has an isomorphism ${}^P({}^P F) \simeq F$; since the canonical morphism

$$H^0(X, F) \longrightarrow H^0(X_t, F|_{X_t}) = 1$$

is not injective, this proves that $({}^P F, f)$ is not cohomologically proper relative to Y in dimension ≤ -1 .

b) Suppose f coherent (i.e. quasi-compact and quasi-separated). Let F be a stack on X . For every geometric point $\bar{y}$ of Y' , we denote by $\bar{Y}$ (resp. $\bar{Y}'$) the strict localization of Y (resp. Y') at $\bar{y}$, and we set $\bar{X} = X \times_Y \bar{Y}$, $\bar{X}' = X' \times_{Y'} \bar{Y}'$, etc. In order that (F, f) be cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1), it is necessary and sufficient that, for every S -scheme S' and for every geometric point $\bar{y}$ of Y' , the canonical functor

$$\bar{F}(\bar{X}) \longrightarrow \bar{F}'(\bar{X}')$$

be faithful (resp. fully faithful, resp. an equivalence).

Indeed, if S' is an S -scheme, in order that the functor

$$g^* f_* F \longrightarrow f'_* F'$$

be faithful (resp. fully faithful, resp. an equivalence), it is necessary and sufficient that the same hold for the functor induced on the fibers at the various geometric points $\bar{y}'$ of $\bar{Y}'$ [2, III 2.1.5.9]. The assertion therefore follows from the computation of the geometric fibers of the direct image of a stack by a coherent morphism [2, VII 2.1.5].

c) Let F be a stack on X . The fact that (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1) is local on Y for the étale topology.

Let S' be an S -scheme, and F' the inverse image of F on X' (cf. (1.0.1)). If (F, f) is cohomologically proper relative to S in dimension ≤ 1 , the same holds for (F', f') . But if (F, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0), the same does not necessarily hold for (F', f') .

For example, let S' be a discrete valuation ring, $f': E_{S'} \rightarrow S'$ the affine space over S' , x a closed point of $E_{S'}$ lying over the generic point of S' , and F' the sheaf of sets on $E_{S'}$ whose restriction to $E_{S'} - \{x\}$ is the constant sheaf with one element and whose fiber at a geometric point over x has two elements. Then (F', f') is not cohomologically proper relative to S' in dimension ≤ -1 . Let $S = S'[Z]$, $f: E_S \rightarrow S$ the affine space over S , and T a closed subset of $X = E_S$ which does not meet the closed set $Z = 0$ and such that $f(T)$ contains the generic point of S . Let G be the inverse image of F' on X and F the sheaf on X obtained by extending $G|_{X - T}$ by the empty set. Then (F, f) is cohomologically proper relative to S in dimension ≤ -1 , but this is no longer so after the change of base $S' \rightarrow S$ defined by $Z = 0$.

d) Let F be a stack on X such that (F, f) is cohomologically proper relative to Y in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1). Then, for every geometric point $\bar{y}$ of Y , the canonical functor

$$(f_*F)_{\bar{y}} \longrightarrow F(X_{\bar{y}})$$

is faithful (resp. fully faithful, resp. an equivalence of categories).

PROPOSITION 1.6. *Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two S -morphisms, and Φ a stack on X .*

- 1) *Suppose that (Φ, f) and $(f_*\Phi, g)$ are cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1). Then the same holds for (Φ, gf) .*
- 2) *Suppose that (Φ, gf) is cohomologically proper relative to S in dimension ≤ -1 (resp. that (Φ, gf) is cohomologically proper relative to S in dimension ≤ 0 and (Φ, f) cohomologically proper relative to S in dimension ≤ -1 , resp. that (Φ, gf) is cohomologically proper relative to S in dimension ≤ 1 and (Φ, f) cohomologically proper relative to S in dimension ≤ 0). Then $(f_*\Phi, g)$ is cohomologically proper relative to S in dimension ≤ -1 (resp. in dimension ≤ 0 , resp. in dimension ≤ 1).*

For every S -scheme S' , we consider the following diagram, all of whose squares are cartesian:

$$(1.6.1) \quad \begin{array}{ccccccc} X' & \xrightarrow{f'} & Y' & \xrightarrow{g'} & Z' & \longrightarrow & S' \\ h \downarrow & & k \downarrow & & m \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y & \xrightarrow{g} & Z & \longrightarrow & S. \end{array}$$

The canonical morphism

$$m^*(g_*f_*\Phi) \longrightarrow g'_*f'_*(h^*\Phi)$$

is identified with the composite of the canonical morphisms

$$m^*(g_*f_*\Phi) \xrightarrow{i} g'_*(k^*f_*\Phi) \xrightarrow{j} g'_*f'_*(h^*\Phi).$$

- 1) The hypothesis implies that i and j are faithful (resp. fully faithful, resp. equivalences); hence the same holds for ji .
- 2) The hypothesis implies that ji is faithful (resp. that ji is fully faithful and j faithful, resp. that ji is an equivalence and j fully faithful); it follows that i is faithful (resp. fully faithful, resp. an equivalence).

COROLLARY 1.7. *Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be two S -morphisms, and let F be a sheaf of groups on X . Suppose that (F, gf) is cohomologically proper relative to S in dimension ≤ -1 (resp. that (F, gf) is cohomologically proper relative to S in dimension ≤ 0 and that (F, f) is cohomologically proper relative to S in dimension ≤ -1). Then (f_*F, g) is cohomologically proper relative to S in dimension ≤ -1 (resp. in dimension ≤ 0).*

Let us take up again the notation of (1.6.1) and, for every scheme Y_1 étale over Y , let us denote by f_1, F_1 the respective inverse images of f, F by the morphism $Y_1 \rightarrow Y$. Let Φ be the stack of torsors under F and Ψ the stack of torsors under f_*F . We have a canonical functor

$$\varphi: \Psi \longrightarrow f_*\Phi,$$

obtained by associating to every scheme Y_1 étale over Y and to every torsor P on Y_1 with group $f_{1*}F_1$ the torsor $\tilde{P}$ on X_1 deduced from f_1^*P by extension of the structure group $f_1^*f_{1*}F_1 \rightarrow F_1$. The functor φ is fully faithful. Indeed, if P and Q are two torsors on Y_1 with group $f_{1*}F_1$, we have a canonical morphism

$$\mathbf{Isom}_{f_{1*}F_1}(P, Q) \longrightarrow f_{1*}(\mathbf{Isom}_{F_1}(\tilde{P}, \tilde{Q}))$$

which is an isomorphism since this is so locally. We deduce from this that the canonical morphism

$$\mathrm{Isom}_{f_{1*}F_1}(P, Q) \longrightarrow \mathrm{Isom}_{F_1}(\tilde{P}, \tilde{Q})$$

is an isomorphism, hence that φ is fully faithful.

We have a commutative diagram

$$\begin{array}{ccc} g'_*k^*\Psi & \xrightarrow{i} & m^*(g_*\Psi) \\ g'_*k^*(\varphi) \downarrow & & \downarrow m^*g_*(\varphi) \\ g'_*k^*(f_*\Phi) & \xrightarrow{j} & m^*(g_*f_*\Phi), \end{array}$$

where i and j are the change of base morphisms. It follows from 1.6 2) that j is faithful (resp. fully faithful). Since $g'_*k^*(\varphi)$ and $m^*g_*(\varphi)$ are fully faithful, we deduce from the above diagram that i is faithful (resp. fully faithful).

COROLLARY 1.8. *Let $f: X \rightarrow Y$ be a coherent S -morphism, $g: Y \rightarrow Z$ a proper S -morphism, and Φ an ind-finite stack on X [2, VII 2.2.1]. Suppose that (Φ, f) is cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1); then the same holds for (Φ, gf) .*

Since f is coherent, $f_*\Phi$ is an ind-finite stack (SGA 4 IX 1.6 (ii)). The corollary therefore follows from 1.6 1) and 1.4.

COROLLARY 1.9. *Let $f: X \rightarrow Y$ be an integral S -morphism, and $g: Y \rightarrow Z$ an S -morphism. If F is a sheaf of sets on X , in order that (f_*F, g) be cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0), it is necessary and sufficient that the same hold for (F, gf) . If F is a sheaf of groups on X , in order that (f_*F, g) be cohomologically proper relative to S in dimension ≤ -1 (resp. ≤ 0 , resp. ≤ 1), it is necessary and sufficient that the same hold for (F, gf) .*

The assertion relating to the case of a sheaf of sets follows from 1.6 and from the fact that (F, f) is cohomologically proper relative to S in dimension ≤ 0 . Let F be a sheaf of groups on X and Φ the stack of torsors under F . By SGA 4 VIII 5.8, every torsor under F is locally trivial over Y . It follows that the stack $f_*\Phi$ is equivalent to the stack of torsors under f_*F , the equivalence being obtained by associating to every scheme Y_1 étale over Y and to every torsor P on $X_1 = X \times_Y Y_1$ with group $F|_{X_1}$ the torsor f_*P with group $f_*F|_{Y_1}$. Since (F, f) is cohomologically proper relative to S in dimension ≤ 1 , the corollary therefore follows from 1.6.

DEFINITIONS 1.10.

1.10.1. Let E be a category and consider a diagram

$$\Phi \xrightarrow{p} \Phi_1 \xrightleftharpoons[p_2]{p_1} \Phi_2,$$

where Φ, Φ_1, Φ_2 are fibered categories over E and the arrows are morphisms of fibered categories, and let

$$a: p_1p \xrightarrow{\sim} p_2p$$

be an isomorphism of functors.

One says that the above diagram is exact if the following condition is satisfied

a) For every pair of objects x, y of Φ and every morphism $f: p(x) \rightarrow p(y)$ such that $p_1(f) = p_2(f)$ (p_1p and p_2p being identified by means of a), there exists a unique morphism $g: x \rightarrow y$ such that $p(g) = f$.

1.10.2. Consider the diagram

$$\Phi \xrightarrow{p} \Phi_1 \xrightarrow{p_1, p_2} \Phi_2 \xrightarrow{p_{12}, p_{23}, p_{31}} \Phi_3 ,$$

where $\Phi, \Phi_i, 1 \leq i \leq 3$, are fibered categories over E and the arrows are morphisms of fibered categories. Suppose given isomorphisms of functors

$$a: p_1 p \xrightarrow{\sim} p_2 p$$

$$a_1: p_{31} p_2 \xrightarrow{\sim} p_{12} p_1, \quad a_2: p_{12} p_2 \xrightarrow{\sim} p_{23} p_1, \quad a_3: p_{23} p_2 \xrightarrow{\sim} p_{31} p_1$$

such that the following diagram is commutative:

$$\begin{array}{ccccc} p_{23} p_1 p & \xrightarrow{\text{id}.a} & p_{23} p_2 p & \xrightarrow{a_3.\text{id}} & p_{31} p_1 p \\ a_2.\text{id} \uparrow & & & & \downarrow \text{id}.a \\ p_{12} p_2 p & \xleftarrow{\text{id}.a} & p_{12} p_1 p & \xleftarrow{a_1.\text{id}} & p_{31} p_2 p \end{array} .$$

One identifies $p_1 p$ and $p_2 p$, $p_{31} p_2$ and $p_{12} p_1$, etc.

One says that the above diagram is exact if the following conditions are satisfied:

a) Analogous to condition a) of 1.10.1.

b) For every object x_1 of Φ_1 and every isomorphism $u: p_1(x_1) \xrightarrow{\sim} p_2(x_1)$ such that

$$(1.10.2.1) \quad p_{23}(u)p_{31}(u) = p_{12}(u)^{-1},$$

there exists an object x of Φ such that there is an isomorphism $i: p(x) \xrightarrow{\sim} x_1$ making the diagram

$$(1.10.2.2) \quad \begin{array}{ccc} p_1 p(x) & = & p_2 p(x) \\ p_1(i) \downarrow & & \downarrow p_2(i) \\ p_1(x_1) & \xrightarrow[u]{\sim} & p_2(x_1) \end{array}$$

commutative.

1.10.3. One defines in the obvious way the notion of a morphism of exact diagrams of fibered categories over a category E .

1.10.4. We shall use in particular the notion of exact diagram in the case where E is a site and $\Phi, \Phi_i, 1 \leq i \leq 3$, are stacks over E .

Let $f: E \rightarrow E'$ be a morphism of sites and let

$$(1.10.4.1) \quad \Phi \xrightarrow{p} \Phi_1 \xrightarrow{p_1, p_2} \Phi_2 \xrightarrow{p_{12}, p_{23}, p_{31}} \Phi_3$$

be an exact diagram of stacks over E . By direct image one obtains a diagram

$$f_* \Phi \longrightarrow f_* \Phi_1 \rightrightarrows f_* \Phi_2 \rightrightarrows f_* \Phi_3$$

which is obviously exact.

If $h: E'' \rightarrow E$ is a morphism of sites, one likewise has an exact diagram

$$h^* \Phi \xrightarrow{p''} h^* \Phi_1 \xrightarrow{p''_1, p''_2} h^* \Phi_2 \xrightarrow{p''_{12}, p''_{23}, p''_{31}} h^* \Phi_3$$

Let us first verify condition a) of 1.10.2. Let F'' be an object of E'' , x'' and y'' two objects of $(h^* \Phi)_{F''}$, x''_1 and y''_1 their respective images in $h^* \Phi_1$, and x''_2, y''_2 their images in $h^* \Phi_2$. Let $u''_1: x''_1 \rightarrow y''_1$ be a morphism such that $p''_1(u''_1) = p''_2(u''_1)$, and let us prove that u''_1 comes from a unique morphism $u'': x'' \rightarrow y''$. The question being local over F'' , one may assume that one has an object F_1 of E , a morphism from F'' into the inverse image F''_1 of F_1 by h , and objects x, y of Φ_{F_1} whose inverse images over F'' are x'' and y'' . Let x_1, y_1 (resp. x_2, y_2) be the images of x, y in Φ_1 (resp. Φ_2). One may assume that u''_1 comes from a morphism

$u_1: x_1 \rightarrow y_1$ such that $p_1(u_1) = p_2(u_1)$. In view of the exactness of (1.10.4.1), one obtains a unique morphism $u: x \rightarrow y$ whose inverse image by h is the required morphism u'' .

Condition b) of 1.10.2 is verified in an analogous way. Let x_1'' be an object of $(h^*\Phi_1)_{F''}$ and $u'': p_1''(x_1'') \rightarrow p_2''(x_1'')$ a morphism satisfying the relation

$$p_{23}''(u'')p_{31}''(u'') = p_{12}''(u'')^{-1},$$

and let us prove that there exist an object x'' of $(h^*\Phi)_{F''}$ and an isomorphism $i'': p''(x'') \simeq x_1''$ making commutative a diagram analogous to (1.10.2.2). Since the question is local over F'' , one may assume that one has an object F_1 , a morphism $F'' \rightarrow F_1''$ as above, and an object x_1 of $(\Phi_1)_{F_1}$ whose inverse image in $(h^*\Phi_1)_{F''}$ is x_1'' . Likewise one may assume that u'' comes from a morphism $u: p_1(x_1) \rightarrow p_2(x_1)$ satisfying (1.10.2.2). The existence of an object x of Φ_{F_1} whose inverse image by h is an element x'' answering the question then follows from the exactness of (1.10.4.1).

EXAMPLES 1.11. 1) Let $f: X_1 \rightarrow X$ be a *descent morphism* for the category of étale sheaves over variable schemes (for example a universally submersive morphism (SGA 4 VIII 9.3)). Let $X_2 = X_1 \times_X X_1$, $g: X_2 \rightarrow X$ the canonical projection, and F a sheaf of sets over X . It then follows from loc. cit. that one has an exact sequence of sheaves of sets

$$(1.11.1) \quad F \longrightarrow f_*f^*F \rightrightarrows g_*g^*F.$$

If Φ is the stack in discrete categories associated to F and Φ_3 the final stack over X , i.e. the stack all of whose fibers are reduced to a single element having as its only morphism the identity morphism, to say that the sequence (1.11.1) is exact amounts to saying that the same holds for the diagram of stacks

$$\Phi \longrightarrow f_*f^*\Phi \rightrightarrows g_*g^*\Phi \rightrightarrows \Phi_3.$$

2) Let $f: X_1 \rightarrow X$ be an *effective descent morphism* for the category of étale sheaves over variable schemes (for example a proper surjective morphism, or an integral surjective morphism, or a faithfully flat morphism locally of finite presentation (SGA 4 VIII 9.4)). Let $X_2 = X_1 \times_X X_1$, $g: X_2 \rightarrow X$ the canonical projection, $X_3 = X_1 \times_X X_1 \times_X X_1$, $h: X_3 \rightarrow X$ the canonical morphism. Let Φ be a stack over X , $\Phi_1 = f_*f^*\Phi$, $\Phi_2 = g_*g^*\Phi$, $\Phi_3 = h_*h^*\Phi$. One then has an exact diagram

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \rightrightarrows \Phi_3,$$

where the arrows are the canonical morphisms associated with the projections.

Indeed, consider Φ as a stack over the category $\mathbf{Sch}_X$ of schemes over X , endowed with the étale topology. Then, by [2, VII 2.2.8], Φ is also a stack for the finest topology on $\mathbf{Sch}_X$ for which the covering morphisms are the effective descent morphisms for the category of étale sheaves. The exactness of the above diagram follows at once.

PROPOSITION 1.12. *Let S be a scheme and $f: X \rightarrow Y$ an S -morphism.*

1) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2$$

be an exact diagram of stacks over X . Suppose that (Φ_1, f) is cohomologically proper relative to S in dimension ≤ 0 and that (Φ_2, f) is cohomologically proper relative to S in dimension ≤ -1 . Then (Φ, f) is cohomologically proper relative to S in dimension ≤ 0 .

2) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \rightrightarrows \Phi_3$$

be an exact diagram of stacks over X . Suppose that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 , that (Φ_2, f) is cohomologically proper relative to S in

dimension ≤ 0 , and that (Φ_3, f) is cohomologically proper relative to S in dimension ≤ -1 . Then (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 .

For every S -scheme S' , one considers the following commutative diagram, all of whose squares are cartesian:

$$\begin{array}{ccccc} X' & \xrightarrow{f'} & Y' & \longrightarrow & S' \\ h \downarrow & & g \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y & \longrightarrow & S \end{array}$$

Let us prove 2), the proof of 1) being analogous. Since the direct image and inverse image functors transform exact diagrams of stacks into exact diagrams (1.10.4), one has the following morphism of exact diagrams of stacks:

$$\begin{array}{ccccccc} g^* f_* \Phi & \xrightarrow{\pi} & g^* f_* \Phi_1 & \xrightarrow[\pi_2]{\pi_1} & g^* f_* \Phi_2 & \rightrightarrows & g^* f_* \Phi_3 \\ \varphi \downarrow & & \varphi_1 \downarrow & & \varphi_2 \downarrow & & \varphi_3 \downarrow \\ f'_* h^* \Phi & \longrightarrow & f'_* h^* \Phi_1 & \rightrightarrows & f'_* h^* \Phi_2 & \rightrightarrows & f'_* h^* \Phi_3 \end{array}$$

By hypothesis φ_1 is an equivalence of categories, φ_2 is fully faithful and φ_3 faithful. It therefore follows from the preceding diagram that φ is an equivalence.

PROPOSITION 1.13. *Let $f: X \rightarrow Y$ be an S -morphism.*

1) *Let*

$$F \longrightarrow G \rightrightarrows H$$

be an exact diagram of sheaves of sets over X . Suppose that (G, f) is cohomologically proper relative to S in dimension ≤ 0 and that (H, f) is cohomologically proper relative to S in dimension ≤ -1 . Then (F, f) is cohomologically proper relative to S in dimension ≤ 0 .

2) *Let $F \rightarrow G$ be a monomorphism of sheaves of groups over X . If Y_1 is a scheme étale over Y , one sets $X_1 = Y_1 \times_Y X$, and one denotes by f_1 (resp. F_1 , resp. G_1) the inverse image of f (resp. F , resp. G) over Y_1 (cf. 1.0.2). Suppose that (G, f) is cohomologically proper relative to S in dimension ≤ 0 (resp. in dimension ≤ 1) and that, for every scheme Y_1 étale over Y and every torsor Q under G_1 , $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ -1 (resp. in dimension ≤ 0). Then (F, f) is cohomologically proper relative to S in dimension ≤ 0 (resp. in dimension ≤ 1).*

3) *Let $F \rightarrow G$ be a monomorphism of sheaves of groups over X . Suppose that (F, f) is cohomologically proper relative to S in dimension ≤ 1 and that (G, f) is cohomologically proper relative to S in dimension ≤ 0 . Then, for every torsor Q under G , $(Q/F, f)$ is cohomologically proper relative to S in dimension ≤ 0 .*

Proof. 1) Let Φ (resp. Φ_1 , resp. Φ_2) be the stack in discrete categories associated to F (resp. G , resp. H) and let Φ_3 be the final stack over X . One then has an exact diagram

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \rightrightarrows \Phi_3.$$

By hypothesis (Φ_1, f) is cohomologically proper relative to S in dimension ≤ 1 and (Φ_2, f) is cohomologically proper relative to S in dimension ≤ 0 (1.2). Since (Φ_3, f) is obviously cohomologically proper relative to S in dimension ≤ -1 , it follows from 1.12 that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 , i.e. that (F, f) is cohomologically proper relative to S in dimension ≤ 0 .

2) Let us first show that, if (G, f) is cohomologically proper relative to S in dimension ≤ 0 and if the $(Q/F_1, f_1)$ are cohomologically proper relative to S in dimension ≤ -1 , then (F, f) is cohomologically proper relative to S in dimension ≤ 0 . By 1.3.1 it suffices to prove

that, for every scheme Y_1 étale over Y and every torsor P over X_1 with group F_1 , $({}^P F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0 when one considers ${}^P F_1$ as a sheaf of sets, and that the canonical morphism

$$d: g^*(R^1 f_* F) \longrightarrow R^1 f'_* F'$$

is injective. The first assertion follows at once from 1), since, if Q denotes the torsor deduced from P by the extension $F_1 \rightarrow G_1$ of the structure group, one has an isomorphism ${}^Q G_1 / {}^P F_1 \xrightarrow{\sim} Q / F_1$.

Let us show that d is injective. It suffices to prove that, if Y_1 is a scheme étale over Y and if P and $\tilde{P}$ are two torsors under F_1 whose inverse images P' and $\tilde{P}'$ over X'_1 are isomorphic, then, after possibly making a surjective étale extension of Y_1 , P and $\tilde{P}$ become isomorphic. Choose an isomorphism $p': P' \xrightarrow{\sim} \tilde{P}'$. Let Q (resp. $\tilde{Q}$) be the torsor deduced from P (resp. $\tilde{P}$) by the extension of the structure group $F_1 \rightarrow G_1$. The inverse images Q' (resp. $\tilde{Q}'$) of Q (resp. $\tilde{Q}$) over X'_1 are deduced from P' (resp. $\tilde{P}'$) by the extension of the structure group $F'_1 \rightarrow G'_1$; let $q': Q' \xrightarrow{\sim} \tilde{Q}'$ be the isomorphism that one obtains in the same way from p' . Since (G, f) is cohomologically proper relative to S in dimension ≤ 0 , one may assume, after possibly making a surjective étale extension of Y_1 , that q' is the image of an isomorphism $q: Q \xrightarrow{\sim} \tilde{Q}$. To the torsor P (resp. $\tilde{P}$) is associated a section x of Q / F_1 (resp. a section $\tilde{x}$ of $\tilde{Q} / F_1$), and, for P and $\tilde{P}$ to be isomorphic, it is necessary and sufficient that there be an isomorphism $Q \xrightarrow{\sim} \tilde{Q}$ such that the isomorphism

$$e: H^0(X_1, Q / F_1) \longrightarrow H^0(X_1, \tilde{Q} / F_1)$$

deduced from it transforms x into $\tilde{x}$. One takes the isomorphism q . The sections $e(x)$ and $\tilde{x}$ of $H^0(X_1, \tilde{Q} / F_1)$ have the same image in $H^0(X'_1, \tilde{Q}' / F'_1)$. Since $(\tilde{Q}' / F'_1, f_1)$ is cohomologically proper relative to S in dimension ≤ -1 , after possibly making a surjective étale extension of Y_1 , one does have $e(x) = \tilde{x}$, which proves the injectivity of d .

To complete the proof, it remains to prove that, if (G, f) is cohomologically proper relative to S in dimension ≤ 1 , and if, for every scheme Y_1 étale over Y and every torsor Q over X_1 with group F_1 , $(Q / F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0 , then the morphism d is surjective. Let P' be a torsor over X'_1 with group F'_1 , and Q' the torsor under G'_1 obtained from P' by extension of the structure group. Giving P' is equivalent to giving Q' and a section x' of $H^0(X'_1, Q' / F'_1)$. It then follows from the surjectivity of the morphism

$$g^*(R^1 f_* G) \longrightarrow R^1 f'_* G'$$

that, after possibly making a surjective étale extension of Y_1 , there exists a torsor Q under G_1 whose inverse image over X'_1 is isomorphic to Q' . Using the fact that $(Q / F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0 , one may likewise assume that there exists an element x of $H^0(X_1, Q / F_1)$ whose image in $H^0(X'_1, Q' / F'_1)$ is x' . The data of Q and of x determine a torsor P under F_1 whose inverse image over X_1 is isomorphic to P' , which proves the surjectivity of d .

3) Let us show that $(Q / F, f)$ is cohomologically proper relative to S in dimension ≤ -1 , i.e. that, for every S -scheme S' and every scheme Y_1 étale over Y , if $x, \tilde{x}$ are two elements of $H^0(X_1, Q_1 / F_1)$ whose images $x', \tilde{x}'$ in $H^0(X'_1, Q'_1 / F'_1)$ are equal, then, after a surjective extension of Y_1 , one has $x = \tilde{x}$. To x (resp. $\tilde{x}$) is associated a torsor P (resp. $\tilde{P}$) under F_1 such that Q_1 is deduced from P (resp. $\tilde{P}$) by the extension $F_1 \rightarrow G_1$ of the structure group. From the relation $x' = \tilde{x}'$ it follows that one has an isomorphism $u': P' \xrightarrow{\sim} \tilde{P}'$ such that the isomorphism induced on Q'_1 by u' is the identity. Since (F, f) is cohomologically proper relative to S in dimension ≤ 0 , one deduces that, after a surjective étale extension of Y_1 , one has an isomorphism $u: P \rightarrow \tilde{P}$ lifting u' ; the fact that (G, f) is cohomologically proper relative to S in dimension ≤ -1 then implies that $x = \tilde{x}$.

Let us show that $(Q / F, f)$ is cohomologically proper relative to S in dimension ≤ 0 . Let Y'' be a scheme étale over Y' and x'' an element of $H^0(X'', Q'' / F'')$. To x'' is associated a

torsor P'' over X'' with group F'' . Since (F, f) is cohomologically proper relative to S in dimension ≤ 1 , one can find surjective étale morphisms $Y_1'' \rightarrow Y''$ and $Y_1 \rightarrow Y$ such that one has a morphism $Y_1'' \rightarrow Y_1'$, and a torsor P over X_1 with group F_1 whose inverse image over X_1'' is isomorphic to the inverse image of P'' . It then follows from the fact that (G, f) is cohomologically proper relative to S in dimension ≤ 0 that one can even choose Y_1'' and Y_1 such that the torsor deduced from P by extension of the structure group $F_1 \rightarrow G_1$ is isomorphic to Q_1 ; to P there corresponds an element x of $H^0(X_1, Q_1/F_1)$ whose image in $H^0(X_1'', Q_1''/F_1'')$ is isomorphic to the inverse image of x'' , which completes the proof.

PROPOSITION 1.14. *Let $f: X \rightarrow S$ be an S -scheme and F a sheaf of sets or of groups over X (resp. a sheaf of ind- $\mathbb{L}$ -groups, where $\mathbb{L}$ is a set of prime numbers). Suppose F locally constant, (F, f) cohomologically proper in dimension ≤ 0 (resp. in dimension ≤ 1), and f locally 0-acyclic (resp. locally 1-aspherical for $\mathbb{L}$) (SGA 4 XV 1.11). Then, for every specialization $\bar{s}_1 \rightarrow \bar{s}_2$ of geometric points of S , the specialization morphism (SGA 4 VIII 7.1)*

$$a_0: (f_*F)_{\bar{s}_2} \longrightarrow (f_*F)_{\bar{s}_1}$$

is an isomorphism, and, if F is a sheaf of groups, the morphism

$$a_1: (R^1 f_*F)_{\bar{s}_2} \longrightarrow (R^1 f_*F)_{\bar{s}_1}$$

is injective (resp. the morphisms a_0 and a_1 are isomorphisms).

The proof is obtained by copying word for word that of SGA 4 XVI 2.3, but replacing in it the expression “proper” by the expression “cohomologically proper”.

COROLLARY 1.15. *Let $f: X \rightarrow S$ be a morphism, Φ a stack over X , and $\mathbb{L}$ a set of prime numbers. Suppose that, for every scheme X_1 étale over X and every pair of objects x, y of Φ_{X_1} , the sheaf $\mathbf{Hom}_{X_1}(x, y)$ is locally constant, that the sheaf $\mathbf{Aut}_{X_1}(x)$ is a locally constant ind- $\mathbb{L}$ -group, and that the sheaf of maximal subgerbes $S\Phi$ of Φ [1, III 2.1.7] is locally constant. Suppose that (Φ, f) is cohomologically proper in dimension ≤ 1 and that f is locally 1-aspherical for $\mathbb{L}$. Then, for every specialization $\bar{s}_1 \rightarrow \bar{s}_2$ of geometric points of S , the specialization morphism*

$$a: (f_*\Phi)_{\bar{s}_2} \longrightarrow (f_*\Phi)_{\bar{s}_1}$$

is an equivalence of categories.

Let $\bar{S}_1$ (resp. $\bar{S}_2$) be the strict localization of S at $\bar{s}_1$ (resp. the strict localization of S at $\bar{s}_2$), $\bar{X}_2, \bar{\Phi}_2$ (resp. $\bar{X}_1, \bar{\Phi}_1$) the inverse images of X_2, Φ_2 over $\bar{S}_2$ (resp. of X_1, Φ_1 over $\bar{S}_1$), and consider the cartesian square

$$\begin{array}{ccc} \bar{X}_1 & \xrightarrow{h} & \bar{X}_2 \\ \bar{f}_1 \downarrow & & \downarrow \bar{f}_2 \\ \bar{S}_1 & \xrightarrow{g} & \bar{S}_2 \end{array}$$

One must show that the functor

$$\varphi: \bar{\Phi}_2(\bar{X}_2) \longrightarrow \bar{\Phi}_1(\bar{X}_1)$$

is an equivalence. The functor φ is fully faithful; indeed, let x, y be two objects of $(\bar{\Phi}_2)_{\bar{X}_2}$; the canonical morphism

$$\mathrm{Hom}_{\bar{X}_2}(x, y) \longrightarrow \mathrm{Hom}_{\bar{X}_1}(\varphi(x), \varphi(y))$$

is identified with the canonical morphism

$$H^0(\bar{X}_2, \mathbf{Hom}_{\bar{X}_2}(x, y)) \longrightarrow H^0(\bar{X}_1, h^*(\mathbf{Hom}_{\bar{X}_2}(x, y))).$$

This morphism is an isomorphism by 1.14.

Let us show that φ is an equivalence. Let x_1 be an object of $\overline{\Phi}_1(\overline{X}_1)$ and G_1 the maximal subgerbe of $\overline{\Phi}_1$ generated by x_1 . The morphism

$$H^0(\overline{X}_2, S\overline{\Phi}_2) \longrightarrow H^0(\overline{X}_1, h^*(S\overline{\Phi}_2)) = H^0(\overline{X}_1, S\overline{\Phi}_1)$$

is bijective, and there therefore exists a maximal subgerbe G_2 of $\overline{\Phi}_2$ such that h^*G_2 is isomorphic to G_1 . It then suffices to prove that the functor

$$G_2 \longrightarrow h_*h^*G_2$$

is an equivalence. But, in this form, the question is local for the étale topology on $\overline{X}_2$. One may therefore assume that G_2 is a gerbe of torsors under the group of automorphisms of an object of G_2 , a case in which the assertion follows from 1.14.

COROLLARY 1.16. *The hypotheses are those of 1.14. If one assumes moreover that F is a sheaf of sets (resp. of ind- $\mathbb{L}$ -groups) and that f_*F (resp. R^1f_*F) is constructible, then f_*F (resp. R^1f_*F) is locally constant.*

The corollary follows from 1.14 by means of SGA 4 IX 2.11.

REMARK 1.17. Let us recall that the condition f locally 0-acyclic is satisfied if f is flat with separable fibers, X and Y locally noetherian (SGA 4 XV 4.1), and that the condition f locally 1-aspherical for $\mathbb{L}$ is satisfied if f is smooth, $\mathbb{L}$ being the set of prime numbers distinct from the residue characteristics of S (SGA 4 XV 2.1).

2. A particular case of cohomological properness: divisors with relative normal crossings

2.0. Let R be a discrete valuation ring with field of fractions K and L an étale K -algebra; L is then a direct product of a finite number of fields L_i , where L_i is an étale extension of K . If L'_i denotes the Galois extension generated by L_i in an algebraic closure of L_i , one says that L is *tamely ramified* over R if the L'_i are tamely ramified extensions in the sense of X 3, i.e. if an inertia group I_i of $L'_i|K$ is of order prime to the residue characteristic p of R .

We know that I_i is in any case an extension of a cyclic group of order prime to p by a p -group. (This follows from [5, ch. IV prop. 7 cor. 4] when the residue extension of R is separable. The proof given in loc. cit. extends to the general case in the following way. Let us take up again the hypotheses and the notation of loc. cit. but without assuming the residue extension separable. Let H_i be the subgroup of the inertia group G_0 , the set of the elements s of G_0 such that one has $s\pi/\pi \in U^i$ for every uniformizer π of A_L . One then checks that G_0/H_1 is a group of order prime to p and that, for $i \geq 1$, the H^i/H^{i+1} are p -groups, from which one deduces the announced result.)

2.0.1. In the case where R is strictly local, one has the following simple characterization: the K -algebra L is tamely ramified over R if and only if the $[L_i : K]$ are prime to p . Moreover, if L is tamely ramified over R , the L_i are cyclic extensions of K . Indeed, when R is strictly local, I_i is equal to the Galois group of L'_i over K . As we have just recalled, I_i is an extension of a cyclic group of order prime to p by a p -group. If one assumes L'_i tamely ramified over R , I_i is then a cyclic group of order prime to p . It follows that $[L_i : K]$ is prime to p and that one has $L_i = L'_i$. Conversely, if $[L_i : K]$ is prime to p , I_i cannot contain a nontrivial normal p -subgroup; I_i is therefore a cyclic group of order prime to p , which proves that L_i is tamely ramified over R .

2.0.2. Let R be a discrete valuation ring with field of fractions K , L an étale K -algebra, and let $\overline{R}$ be a strict localization of R , $\overline{K}$ its field of fractions, $\overline{L} = L \otimes_K \overline{K}$. Then, for L to be tamely ramified over R , it is necessary and sufficient that $\overline{L}$ be tamely ramified over $\overline{R}$. Indeed one reduces to the case where L is a field. Let then $\overline{L} = \prod_i \overline{L}_i$, where the $\overline{L}_i$ are fields, extensions of $\overline{K}$; if L' is the Galois extension generated by L , and, if $\overline{L}' = L' \otimes_K \overline{K}$, one likewise has a decomposition of $\overline{L}'$ into a product of fields, $\overline{L}' = \prod_j \overline{L}'_j$ and each $\overline{L}_i$ is a subextension of at least one of the $\overline{L}'_j$. Since L' is a Galois extension of K , the $\overline{L}'_j$ are

Galois extensions of $\bar{K}$. Suppose L tamely ramified over R ; since the Galois group of $\bar{L}_j|K$ is isomorphic to the inertia group of $L'|K$, the L'_j are also tamely ramified over $\bar{R}$, and hence so are the $\bar{L}_i$. Conversely, suppose $\bar{L}$ tamely ramified over $\bar{R}$. For each j , let v_j be the discrete valuation of $\bar{L}_j$ which extends the valuation of $\bar{K}$, and let us still denote by v_j the valuation induced on L' . As j varies, v_j runs through the set of valuations of L which extend the valuation of K . Let $G = \text{Gal}(L'|K)$, $H = \text{Gal}(L'|L)$, I_j the inertia group of $L'|K$ at v_j , J_j the inertia group of $L'|L$ at v_j . The group I_i is an extension of a cyclic group of order prime to p by a p -group P_j . Since the $\bar{L}_i$ are tamely ramified over $\bar{R}$, I_j/J_j is of order prime to p , hence one has $P_j \subset J_j$. Consequently the group H contains all the P_j , hence also the group P generated by the P_j for variable j . But the group P is invariant in G , since an inner automorphism of G transforms the I_j into one another, hence also the P_j into one another. It follows that P is a subgroup of H normal in G , hence, since L' is the Galois extension generated by L , that one has $P = 1$, which proves that L is tamely ramified over R .

More generally, let $R \rightarrow R'$ be a morphism of discrete valuation rings such that the image of a uniformizer π of R is a uniformizer π' of R' and such that the residue extension $k(R')$ is a separable extension of $k(R)$. Let K be the field of fractions of R , K' the field of fractions of R' , L an étale K -algebra, $L' = L \otimes_K K'$. Then, for L to be tamely ramified over R , it is necessary and sufficient that L' be tamely ramified over R' . Indeed we may assume R and R' strictly local. By 2.0.1 it suffices to prove that, when L is a field, so is L' . Let $\tilde{R}$ be the normalization of R in L , $\tilde{\pi}$ a uniformizer of $\tilde{R}$, $\tilde{R}' = \tilde{R} \otimes_R R'$. The extension $k(\tilde{R})|k(R)$ being radicial and the extension $k(R')|k(R)$ étale, $k(\tilde{R}) \otimes_{k(R)} k(R')$ is a field [EGA IV 4.3.2 and 4.3.5]. This proves that R' is a local ring, and, since π has image π' in R' , one has $k(\tilde{R}') = R' / (\tilde{\pi})$; consequently R' is a discrete valuation ring [5, ch. I §2 prop. 2], hence L' is a field.

2.0.3. By reduction to the strictly local case, one sees that a subalgebra of a tamely ramified algebra is tamely ramified, that the tensor product of two tamely ramified algebras is tamely ramified, that a tamely ramified algebra remains so after extension of the discrete valuation ring, that an algebra which becomes tamely ramified after a tamely ramified extension is tamely ramified.

2.1. Let X be an S -scheme, D a divisor ≥ 0 on X . Recall (SGA 5 II 4.2) that one says that D has strictly *normal crossings relative to S* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma(X, \mathcal{O}_X)$ such that one has $D = \sum_{i \in I} \text{div}(f_i)$ and such that the following condition is satisfied:

2.1.0. For every point x of $\text{Supp } D$, X is smooth over S at x , and, if one denotes by $I(x)$ the set of the $i \in I$ such that $f_i(x) = 0$, the subscheme $V((f_i)_{i \in I(x)})$ is smooth over S of codimension $\text{card } I(x)$ in X .

The divisor D is said to have *normal crossings relative to S* if, locally on X for the étale topology, it has strictly normal crossings.

Let D be a divisor with normal crossings relative to S . We set $Y = \text{Supp } D$, $U = X - Y$, and we denote by $i: U \rightarrow X$ the canonical immersion. For every geometric point $\bar{s}$ of S and for every maximal point y of the geometric fiber $Y_{\bar{s}}$, the ring $R = \mathcal{O}_{X_{\bar{s}}, y}$ is a discrete valuation ring.

In the rest of this no., we shall use the following technical definition:

DEFINITION 2.1.1. Let F be a sheaf of sets on U . One says that F is *tamely ramified over X (along D) relative to S* if, for every geometric point $\bar{s}$ of S , the following condition is satisfied:

For every maximal point y of $Y_{\bar{s}}$, the restriction of F to the field of fractions K of $\mathcal{O}_{X_{\bar{s}}}$ is representable by the spectrum of an étale K -algebra L , tamely ramified over $\mathcal{O}_{X_{\bar{s}}, y}$.

Most often, when no confusion can result from it, we shall omit the mention of D in the terminology.

DEFINITION 2.1.2. If F is a sheaf of groups on U , tamely ramified over X relative to S , one denotes by

$$H_t^1(U, F)$$

the subset of $H^1(U, F)$ formed by the classes of torsors under F which are tamely ramified over X relative to S .

Let

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ g \downarrow & \swarrow f & \\ T & & \end{array}$$

be a commutative diagram of S -schemes, with i as in 2.1; one denotes by

$$R_t^1 g_* F$$

the sheaf on T associated with the presheaf $T' \mapsto H_t^1(U', F)$, where T' runs through the schemes étale over T and where $U' = U \times_T T'$; $R_t^1 g_* F$ is a subsheaf of $R^1 g_* F$.

Note that, if g is coherent, if $\bar{t}$ is a geometric point of T , $\bar{T}$ the strict localization of T at $\bar{t}$, $\bar{U} = U \times_T \bar{T}$, one has an isomorphism

$$(2.1.2.1) \quad (R_t^1 g_* F)_{\bar{t}} \simeq H_t^1(\bar{U}, \bar{F})$$

2.1.3. Let $C_t((U, X)/S)$, or simply C_t , be the category of étale coverings of U which are tamely ramified over X relative to S . Suppose U connected and let a be a geometric point of U . Let Γ_t be the functor which, to an étale covering U' of U tamely ramified over X relative to S , associates the set of the geometric points of U' over a . From 2.0 it follows that the pair (C_t, Γ_t) satisfies axioms (G_1) to (G_6) of V 4. Consequently Γ_t is representable by a pro-object which is called the *tamely ramified universal covering of (U, X) relative to S pointed at a* . The group opposite to the group of U -automorphisms of the tamely ramified universal covering is called the *tamely ramified fundamental group* and denoted

$$\pi_1^t((U, X)/S, a) \text{ or simply } \pi_1^t(U, a) \text{ or even } \pi_1^t(U).$$

It is obviously a quotient of the fundamental group $\pi_1(U, a)$ (V 6.9).

2.1.4. Let F be a sheaf of groups on U , P a right torsor with group F , Q a left torsor with group F , and suppose P and Q tamely ramified over X relative to S . Then so is $P \overset{F}{\wedge} Q$. Indeed one reduces to showing that, if R is a discrete valuation ring with field of fractions K and if F is a finite étale group scheme over K , and if P and Q are two torsors under F tamely ramified over R , then so is $P \overset{F}{\wedge} Q$. Now $T = P \overset{F}{\wedge} Q$ is a quotient of $P \times_K Q$. If L, M, N denote the K -algebras representing T, P, Q respectively, then L is a subalgebra of $M \otimes_K N$, and it follows from 2.0.3 that L is tamely ramified over R .

We deduce from the preceding that, if F is a sheaf of groups on U and if there exists a torsor with group F tamely ramified over X relative to S , then F is tamely ramified over X relative to S . Indeed the torsor P° opposite to P is tamely ramified over X relative to S , since it is isomorphic to P as a sheaf of sets. If ${}^P F$ is the group obtained by twisting F by P , one has an isomorphism

$$F \simeq P^\circ \overset{{}^P F}{\wedge} P$$

and consequently F is tamely ramified over X relative to S .

One sees as before that, if $F \rightarrow F'$ is a morphism of sheaves of groups on U , tamely ramified over X relative to S , and if P is a torsor under F tamely ramified over X relative to S , then the torsor P' deduced from P by the extension of the structure group $F \rightarrow F'$ is tamely ramified over X relative to S .

In particular the canonical morphism

$$H^1(U, F) \longrightarrow H^1(U, F')$$

gives by restriction to $H_t^1(U, F)$ a canonical morphism

$$H_t^1(U, F) \longrightarrow H_t^1(U, F')$$

2.1.5. Let $S' \rightarrow S$ be a morphism and let us denote by U' (resp. X' , etc.) the inverse image of U (resp. X , etc.) over S' . If F is a sheaf of sets on U tamely ramified over X relative to S , it follows from definition 2.1.1 and from 2.0.3 that F' is tamely ramified over X' relative to S' .

If now F is a sheaf of groups on U , the inverse image over S' of a torsor under F tamely ramified over X relative to S is a torsor under F' tamely ramified over X' relative to S' . In particular one has a canonical functor

$$(2.1.5.1) \quad C_t((U, X)/S) \longrightarrow C_t((U', X')/S').$$

Suppose U and U' connected and let a be a geometric point of U , a' a geometric point of U' over a ; one deduces from the preceding a canonical morphism

$$(2.1.5.2) \quad \pi_1^t(U', a') \longrightarrow \pi_1^t(U, a).$$

If $S' \rightarrow S$ is a morphism and $h: T' \rightarrow T$ the canonical projection, the morphism

$$h^*(R^1 g_* F) \longrightarrow R^1 g'_* F'$$

gives by restriction a canonical morphism

$$(2.1.5.3) \quad h^*(R_t^1 g_* F) \longrightarrow R_t^1 g'_* F'$$

2.1.6. Let F be a sheaf of groups on U , tamely ramified over X relative to S . The notation being that of 2.1.2, one has canonical exact sequences:

$$(2.1.6.1) \quad \begin{aligned} 1 &\longrightarrow H^1(X, i_* F) \longrightarrow H_t^1(U, F) \longrightarrow H^0(X, R_t^1 i_* F) \\ 1 &\longrightarrow R^1 f_*(i_* F) \longrightarrow R_t^1 g_* F \longrightarrow f_*(R_t^1 i_* F). \end{aligned}$$

The first is obtained from the exact sequence (SGA 4 III 3.2):

$$1 \longrightarrow H^1(X, i_* F) \longrightarrow H^1(U, F) \longrightarrow H^0(X, R^1 i_* F).$$

Indeed it suffices to show that the image of $H^1(X, i_* F)$ in $H^1(U, F)$ is in fact contained in $H_t^1(U, F)$ and that the image of $H_t^1(U, F)$ in $H^0(X, R^1 i_* F)$ is in fact contained in $H^0(X, R_t^1 i_* F)$. Now the inverse image over U of a torsor under $i_* F$ is a torsor under $i^* i_* F$ which is obviously tamely ramified over X relative to S , hence the same holds after the extension of the structure group $i^* i_* F \rightarrow F$, which proves the existence of the arrow $H^1(X, i_* F) \rightarrow H_t^1(U, F)$. The fact that the image of $H_t^1(U, F)$ in $H^0(X, R^1 i_* F)$ is contained in $H^0(X, R_t^1 i_* F)$ follows immediately from the definition of $R_t^1 i_* F$. This proves the existence of the first exact sequence, and the second is deduced from it by localization.

2.2. We keep the notation of 2.1. We are going to define a notion of tamely ramified object of a stack Φ on U when the latter is given, locally on X and on S for the étale topology, as the *inverse image of a stack Ψ on S* .

First let G be a gerbe on U and suppose given an étale surjective morphism $S_1 \rightarrow S$, an étale surjective morphism $X_2 \rightarrow X \times_S S_1$, a trivial gerbe H on S_1 and an isomorphism

$$G|_{U_2} \longrightarrow H|_{U_2},$$

where $U_1 = U \times_X X_1$, $U_2 = U \times_X X_2$. When one chooses a trivialization of $H|_{X_2}$, the isomorphism above identifies $G|_{U_2}$ with the stack of torsors under a sheaf of groups F . One says that an element x of G_U is tamely ramified over X relative to S if the restriction of x to U_2 is a torsor tamely ramified over X relative to S . By 2.1.4 this notion does not depend on the way in which one has trivialized $H|_{X_2}$.

Now let Φ be a stack on U and suppose given an étale surjective morphism $S_1 \rightarrow S$, an étale surjective morphism $X_2 \rightarrow X \times_S S_1$, a stack Ψ on S_1 and an isomorphism

$$i: \Phi|_{U_2} \longrightarrow \Psi|_{U_2}.$$

Let x be an element of Φ_U , G_x the maximal subgerbe of Φ generated by x [1, III 2.1.7], $S\Phi$ the sheaf of maximal subgerbes of Φ . The isomorphism i induces an isomorphism

$$S\Phi|_{U_2} \longrightarrow S\Psi|_{U_2}.$$

It follows from 5.7 that, after replacing S_1 if necessary by an étale surjective extension, one has a unique maximal subgerbe H of Ψ , which one may assume trivial, such that i defines an isomorphism

$$G_x|_{U_2} \longrightarrow H|_{U_2}.$$

One says that the element x is tamely ramified over X relative to S if it is so as an element of G_x endowed with the isomorphism above.

2.2.1. Let Φ be a stack on U given, locally on X and S , as the inverse image of a stack on S , and let

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ g \downarrow & \swarrow f & \\ T & & \end{array}$$

be a diagram as in 2.1.2. For every scheme T' étale over T , if $U' = U \times_T T'$, one considers the subset $(g_*^t \Phi)_{T'}$ of $(g_* \Phi)_{T'} = \Phi_{U'}$ formed by the elements of $\Phi_{U'}$ which are tamely ramified over X relative to S . One calls tamely ramified direct image of Φ by g , and one denotes by

$$g_*^t \Phi$$

the full subcategory of $g_* \Phi$ whose objects over a scheme T' étale over T are the elements of $(g_*^t \Phi)_{T'}$. It is clear that $g_*^t \Phi$ is a substack of $g_* \Phi$.

2.2.2. By reduction to the case of a stack of torsors, one sees that, if Φ is a stack on U which is, locally for the étale topology of S and X , the inverse image of a stack on S , the canonical morphism

$$h^*(g_* \Phi) \longrightarrow g'_* \Phi'$$

gives by restriction a canonical morphism

$$h^*(g_*^t \Phi) \longrightarrow g'^t \Phi'$$

REMARKS 2.3. a) If F is a locally constant constructible sheaf of sets on U , for F to be tamely ramified over X relative to S , it suffices that the condition of 2.1.1 be satisfied for the geometric points of S over the maximal points of S . To see this we may assume that the divisor D has strictly normal crossings. The sheaf F is representable by an étale covering V of U . If $\bar{s}$ is a geometric point of S , y a maximal point of $Y_{\bar{s}}$, we denote by $\bar{S}$ the strict localization of S at $\bar{s}$, $\bar{X}$ the strict localization of X at $\bar{y}$, $\bar{U} = U \times_X \bar{X}$, $\bar{V} = V \times_X \bar{X}$. If the condition of 2.1.1 is satisfied at the geometric points over the maximal points of S , it follows from 5.5 below that $\bar{V}$ is a covering of $\bar{U}$ tamely ramified over $\bar{X}$ relative to $\bar{S}$; consequently V is an étale covering of U tamely ramified over X relative to S .

b) Let F be a sheaf of groups on U tamely ramified over X relative to S . If $\bar{s}$ is a geometric point of S , y a maximal point of $Y_{\bar{s}}$, we denote by K the field of fractions of $\mathcal{O}_{X_{\bar{s}}, y}$. Suppose that, for every point $\bar{s}$ and for every point y , the K -algebra L whose spectrum represents $F|_K$ is of rank prime to the residue characteristic p of $\mathcal{O}_{X_{\bar{s}}}$. We shall sometimes say, by abuse of language, that F is prime to the residue characteristics of S . When this is so, every torsor P under F is tamely ramified over X relative to S .

Indeed, let $\bar{R}$ be the strict localization of $\mathcal{O}_{X_{\bar{s}}, y}$ at $\bar{y}$, $\bar{K}$ its field of fractions, $\bar{F}$ the inverse image of F on $\bar{K}$. Let us show that we may assume F constant. Since F is tamely ramified over X relative to S , $\bar{F}$ is representable by the spectrum of a $\bar{K}$ -algebra $L = \prod L_i$, where the L_i are extensions of $\bar{K}$ of degree prime to p . One can therefore find an extension K' of $\bar{K}$ of degree prime to p such that $\bar{F}|_{K'}$ is a constant sheaf. By 2.0.3, to prove that $P|_{\bar{K}}$ is tamely ramified over $\bar{R}$, it suffices to see that $P|_{K'}$ is tamely ramified over the integral closure of $\bar{R}$ in K' , whence the reduction to the case where $\bar{F}$ is constant. Suppose henceforth $\bar{F}$ constant.

The $\overline{K}$ -algebra H which represents $P|\overline{K}$ is then a product of extensions H_i of $\overline{K}$ isomorphic to one another. Since the rank of H is prime to p , so is $[H_1 : \overline{K}]$, which proves that H is tamely ramified over X relative to S .

c) Let X be a regular scheme, D a divisor with normal crossings of X (SGA 5 I 3.1.5), $U = X - \text{Supp } D$, F a sheaf of sets on U . If y is a maximal point of $\text{Supp } D$, we denote by K the field of fractions of $\mathcal{O}_{X,y}$. One says that F is *tamely ramified relative to D* if, for every maximal point y of $\text{Supp } D$, $F|K$ is representable by a K -algebra tamely ramified over $\mathcal{O}_{X,y}$.

THEOREM 2.4. *Let $f: X \rightarrow S$ be an S -scheme, D a divisor on X with normal crossings relative to S (2.1), $Y = \text{Supp } D$, $U = X - Y$, $i: U \rightarrow X$ the canonical immersion. Let F be a sheaf of sets (resp. of groups) on U , satisfying one of the following conditions:*

- a) *F is, locally for the étale topology on X and on S , the inverse image of a sheaf of sets (resp. of a constructible sheaf of groups) on S .*
- b) *F is locally constant constructible on U and tamely ramified over X relative to S .*

Then we have the following conclusions:

- 1) *(F, i) is cohomologically proper relative to S in dimension ≤ 0 (resp. for every morphism $h: S' \rightarrow S$, if $i': U' \rightarrow X'$ is the inverse image of i over S' , if $F' = F|U'$ and if $k = h_{(X)}$, the canonical morphism*

$$\Psi: k^*(R_{\mathfrak{t}}^1 i_* F) \longrightarrow R_{\mathfrak{t}}^1 i'_* F'$$

is an isomorphism).

If F is a sheaf of groups prime to the residue characteristics of S (2.3.b) (tamely ramified over X relative to S), then (F, i) is cohomologically proper relative to S in dimension ≤ 1 .

- 2) *If F is a constructible sheaf of sets (resp. of groups), $i_* F$ (resp. $R_{\mathfrak{t}}^1 i_* F$) is constructible.*

Proof. For every S -scheme S' , we consider the following diagram, all of whose squares are cartesian:

$$\begin{array}{ccccc} U & \xleftarrow{\quad} & U' & & \\ \downarrow g & \searrow i & \downarrow g' & \searrow i' & \\ & X & \xleftarrow{k} & X' & \\ \downarrow f & & \downarrow f' & & \\ S & \xleftarrow{h} & S' & & \end{array}$$

Since the question is local on X for the étale topology, we may assume that D is a divisor with strictly normal crossings relative to S (2.1); moreover, after restricting X to a neighborhood of Y , we may assume X smooth over S .

Proof of 2.4 1)

2.4.1. *Case of a sheaf of sets satisfying a).* We may assume that we have $F = g^*G$, where G is a sheaf on S . It then follows from (SGA 4 XVI 3.2) that the canonical morphism

$$(2.4.1.1) \quad f^*G \longrightarrow i_* F$$

is an isomorphism. For every S -scheme $h: S' \rightarrow S$, we have likewise an isomorphism $f'^*G' \rightarrow i'_* F'$; consequently the canonical morphism

$$\varphi: k^*(i_* F) \longrightarrow i'_* F'$$

is identified with the natural isomorphism

$$k^* f^* G \simeq f'^* G'$$

2.4.2. *Case of a sheaf of sets satisfying b).* We must show that φ is an isomorphism, and for this it suffices to see that this is so at each geometric point $\bar{x}'$ of X' . Let $\bar{S}$ (resp. $\bar{X}$, resp. $\bar{S}'$, resp. $\bar{X}'$) be the strict localization of S (resp. X , resp. S' , resp. X') at $\bar{x}'$ and let us put $\bar{U} = U_{(\bar{X})}$, $\bar{U}' = U_{(\bar{X}')}$, etc. The morphism $\varphi_{\bar{x}}$ is identified with the canonical morphism

$$\bar{\varphi}: H^0(\bar{U}, \bar{F}) \longrightarrow H^0(\bar{U}', \bar{F}')$$

We can find a principal covering V of $\bar{U}$, of the type appearing in 5.4, such that the inverse image of $\bar{F}$ on V is a constant sheaf with value C . If Π is the Galois group of V over $\bar{U}$, Π operates on $\bar{F}|V$, and we have

$$(2.4.2.1) \quad H^0(\bar{U}, \bar{F}) \simeq H^0(V, C_V)^\Pi$$

where the right-hand side denotes the set of the elements of $H^0(V, C_V)$ invariant under Π . Since $V' = V \times_{\bar{U}} \bar{U}'$ is a principal covering of $\bar{U}'$ with Galois group $\Pi' \simeq \Pi$, we see that the morphism $\bar{\varphi}$ is obtained, by taking the invariants under Π , from the canonical morphism

$$H^0(V, C_V) \longrightarrow H^0(V', C_{V'}).$$

Since V and V' are connected (5.4), this morphism, hence also $\bar{\varphi}$, is an isomorphism.

Let us note that if moreover F is a sheaf of groups and if P is a torsor on $\bar{U}$ with group $\bar{F}$, tamely ramified relative to D , it follows from the preceding proof and from 2.2 that $({}^P\bar{F}, \bar{i})$ is cohomologically proper relative to S in dimension ≤ 0 .

2.4.3. *Case of a sheaf of groups.* To show that Ψ is an isomorphism, it suffices to prove that, for every geometric point $\bar{y}'$ of Y' , the morphism

$$\Psi_{\bar{y}'}: (k^*(R_t^1 i_* F))_{\bar{y}'} \longrightarrow (R_t^1 i'_* F')_{\bar{y}'}$$

is an isomorphism. Now, by 2.1.2, $\Psi_{\bar{y}'}$ is identified with the canonical morphism

$$\bar{\Psi}: H_t^1(\bar{U}, \bar{F}) \longrightarrow H_t^1(\bar{U}', \bar{F}').$$

Let $\tilde{U}$ be the universal tamely ramified covering of $\bar{U}$ (2.1.3) and $\tilde{F}$ the inverse image of $\bar{F}$ on $\tilde{U}$. It follows from 5.7 in case a) and from 5.5 in case b) that $H_t^1(\bar{U}, \bar{F})$ is identified with the subset $H^1(\bar{U}, \bar{F})$ formed by the classes of $\bar{F}$ -torsors whose inverse image on $\tilde{U}$ is trivial. On the other hand a classical argument (cf. IX 5, p. page-300) shows that the set of the elements of $H^1(\bar{U}, \bar{F})$ whose inverse image on $\tilde{U}$ is trivial is identified with $H^1(\pi_1^t(\bar{U}), H^0(\tilde{U}, \tilde{F}))$. We thus obtain a canonical isomorphism

$$(2.4.3.1) \quad H_t^1(\bar{U}, \bar{F}) \xrightarrow{\sim} H^1(\pi_1^t(\bar{U}), H^0(\tilde{U}, \tilde{F})).$$

Consequently the morphism $\bar{\Psi}$ is identified with the canonical morphism

$$H^1(\pi_1^t(\bar{U}), H^0(\tilde{U}, \tilde{F})) \longrightarrow H^1(\pi_1^t(\bar{U}'), H^0(\tilde{U}', \tilde{F}')).$$

Let us show that this morphism is an isomorphism. The morphism $\pi_1^t(\bar{U}') \rightarrow \pi_1^t(\bar{U})$ is an isomorphism by 5.6, and the same holds for the morphism $H^0(\tilde{U}, \tilde{F}) \rightarrow H^0(\tilde{U}', \tilde{F}')$. Indeed, this is obvious in case b) since $\tilde{F}$ is constant and $\tilde{U}$ and $\tilde{U}'$ are connected. In case a), let $\bar{G}$ be a constructible sheaf of groups on $\bar{S}$ such that we have $\bar{F} = \bar{g}^* \bar{G}$; since the morphisms $\tilde{U} \rightarrow \bar{S}$ and $\tilde{U}' \rightarrow \bar{S}'$ are 0-acyclic (5.7), we have

$$H^0(\tilde{U}, \tilde{F}) \simeq H^0(\bar{S}, \bar{G}) \simeq H^0(\bar{S}', \bar{G}') \simeq H^0(\tilde{U}', \tilde{F}'),$$

which implies that Ψ is an isomorphism. The last assertion of 2.4 1) follows from what precedes, taking into account 2.3 b).

Proof of 2.4 2). The case of a constructible sheaf of sets satisfying a) follows at once from (2.4.1.1). Let $F = g^*G$ be a sheaf of groups satisfying a), where G is a constructible sheaf; we may assume S affine; let $(S_j)_{j \in J}$ be a finite family of reduced closed subschemes of S whose union covers S , such that the inverse image of G on S_j is a locally constant sheaf. Taking into account 2.4 1), it suffices, in order to establish that $R_t^1 i_* F$ is constructible, to see that this is so after the change of base $S_j \rightarrow S$, for each $j \in J$. We are thus reduced to case b) where F is locally constant.

We assume from now on that F is a sheaf of sets or of groups satisfying b). Since the question is local for the étale topology on X , we may assume X of finite presentation over S , and, by passage to the limit, we may assume X and S noetherian.

Let $D = \sum_{1 \leq i \leq r} \text{div } f_i$, where, for each point x of $\text{Supp } D$, if $I(x)$ is the set of the i such that $f_i(x) = 0$, the subscheme $V((f_i)_{i \in I(x)})$ is smooth over S of codimension $\text{card } I(x)$ in X . Let $\mathcal{P}$ be the set of subsets of $[1, r]$ and, for each $I \in \mathcal{P}$, let us put

$$X_I = \left(\bigcap_{i \in I} V(f_i) \right) \cap \left(\bigcap_{i \notin I} X_{f_i} \right).$$

Let z be a point of X_I . After first restricting to an étale neighborhood of z , we can find an open subset W of X containing z and a principal covering V of $U \cap W$, tamely ramified over W relative to S , of the type considered in 5.6.1, such that the inverse image of F on V is a constant sheaf with value C . Let π be the Galois group of the covering V . For every geometric point $\bar{x}$ of X_I , we then have, by (2.4.2.1),

$$H^0(\bar{U}, \bar{F}) \simeq H^0(V, C_V)^\pi.$$

It follows that $i_* F|_{X_I \cap W}$ is locally constant (SGA 4 IX 2.13) and consequently $i_* F$ is constructible.

Let us finally show that if F is a locally constant sheaf of groups, $R_t^1 i_* F|_{X_I}$ is constructible. If $\bar{x}$ is a geometric point of X , we have obtained in (2.4.3.1) the expression

$$H_t^1(\bar{U}, \bar{F}) \xrightarrow{\sim} H^1(\pi_1^t(\bar{U}), H^0(\tilde{U}, \tilde{F})).$$

If p is the residue characteristic of $\bar{X}$, we have, by 5.6, $\pi_1^t(\bar{U}) = \prod_{l \neq p} \mathbf{Z}_l(1)^{\text{card } I}$. Let us denote by $\mathbb{L}$ the set of prime numbers which divide the order of the finite group $H^0(\tilde{U}, \tilde{F})$ and let $K = \prod_{l \in \mathbb{L} - \{p\} \cap \mathbb{L}} \mathbf{Z}_l(1)^{\text{card } I}$. It follows from [4, I § 5 ex.2] that we have

$$H_t^1(\bar{U}, \bar{F}) \simeq H^1(K, H^0(\tilde{U}, \tilde{F})).$$

Since K is topologically of finite type and $H^0(\tilde{U}, \tilde{F})$ finite, we first of all deduce from this that the fibers of the sheaf $R_t^1 i_* F|_{X_I}$ are finite. On the other hand, the set $\mathbb{L}$ does not depend on the point $\bar{x}$. For every $q \in \mathbb{L}$, let $X_{I,q}$ be the closed subset of X_I with equation $q = 0$ and let $X_{I'}$ be the open subset of X_I complementary to the union of the $X_{I,q}$. Then $R_t^1 i_* F|_{X_{I,q}}$ and $R_t^1 i_* F|_{X_{I'}}$ are locally constant; indeed a specialization arrow of geometric points of $X_{I,q}$ (resp. of $X_{I'}$) induces an isomorphism on the groups K (5.6.1) hence also on the sets $H_t^1(\bar{U}, \bar{F})$, and one can apply SGA 4 IX 2.13.

COROLLARY 2.5. *Let $f: X \rightarrow S$ be a morphism, D a divisor on X with normal crossings relative to S (2.1), $Y = \text{Supp } D$, $U = X - Y$, $i: U \rightarrow X$ the canonical immersion. Let Φ be a stack on U and suppose given surjective étale morphisms $S_1 \rightarrow S$ and $X_2 \rightarrow X \times_S S_1$, a stack Ψ on S_1 and an isomorphism $\Phi|_{U_2} \simeq \Psi|_{U_2}$ (cf. 2.2).*

Then, for every morphism $h: S' \rightarrow S$, if $k = h_{(X)}$, the canonical functor

$$\varphi: k^* i_* \Phi \longrightarrow i'_* \Phi'$$

is fully faithful. If Ψ is constructible, the canonical functor

$$\psi: k^* i_*^t \Phi \longrightarrow i_*'^t \Phi'$$

is an equivalence of categories.

Moreover, if the stack Ψ is constructible (resp. if Ψ is 1-constructible (0) and S locally noetherian), $i_*\Phi$ is constructible (resp. $i_*^t\Phi$ is 1-constructible).

Let us show that φ is fully faithful. It suffices to see that, for every geometric point $\bar{x}'$ of X' , this is so for the functor

$$\bar{\varphi}: \Phi(\bar{U}) \longrightarrow \Phi'(\bar{U}')$$

(we have taken up again the notation of 2.4.2). Let a, b be two elements of $\Phi(\bar{U})$, a', b' their images under $\bar{\varphi}$. Since the morphism $\bar{U} \rightarrow \bar{S}$ is locally 0-acyclic (5.7), we have an isomorphism

$$H^0(\bar{U}, \bar{S}\Phi) \xrightarrow{\sim} H^0(S, \bar{S}\Psi).$$

Consequently a and b come by inverse image from elements of Ψ , and hence so does $F = \mathbf{Hom}_{\bar{U}}(a, b)$. Since the sheaf $F' = \mathbf{Hom}_{\bar{U}'}(a', b')$ is the inverse image on $\bar{U}'$ of F , it follows from 2.4.1 that the canonical morphism

$$H^0(\bar{U}, F) \longrightarrow H^0(\bar{U}', F')$$

is an isomorphism, which proves that $\bar{\varphi}$ is fully faithful.

Let us show that, for every geometric point $\bar{x}'$ of X' , the functor

$$\bar{\psi}: i_*^t\Phi(\bar{X}') \longrightarrow i_*'^t\Phi'(\bar{X}')$$

is an equivalence. By what precedes, $\bar{\psi}$ is fully faithful. Let us show that $\bar{\psi}$ is essentially surjective. Let a' be an element of $\Phi'(\bar{U}')$ tamely ramified over X' relative to S' (2.2) and let us show that it is the image of a tamely ramified element of $\Phi(\bar{U})$. It follows from 2.4 1) that the canonical morphism

$$H^0(\bar{U}, \bar{S}\Phi) \longrightarrow H^0(\bar{U}', \bar{S}\Phi')$$

is an isomorphism. Let G' be the maximal subgerbe of Φ' generated by a' ; there then exists a maximal subgerbe G of Φ , inverse image of a gerbe on $\bar{S}$, such that we have

$$\bar{m}^*G \simeq G',$$

where m is the morphism $\bar{U}' \rightarrow \bar{U}$. The canonical functor

$$(*) \quad \bar{k}^* i_*^t G \longrightarrow i_*'^t G'$$

is an equivalence, since G is identified with a gerbe of torsors under a constructible sheaf of groups coming from $\bar{S}$, and one can apply 2.4 1). It then follows from (*) that there exists an element a of $G(\bar{U})$, tamely ramified over X relative to S , whose inverse image on $\bar{U}'$ is a' , which proves that $\bar{\psi}$ is an equivalence.

If Ψ is constructible, so is $i_*\Phi$; indeed an object x of $i_*\Phi$ is, locally for the étale topology of S , the inverse image of an object y of Ψ ; it therefore follows from 2.4.1.1 that $\mathbf{Aut}(x)$ is the inverse image of $\mathbf{Aut}(y)$, hence is constructible. Finally, if Ψ is 1-constructible, so is $i_*^t\Phi$ by 6.3 below.

COROLLARY 2.6. *The notation is that of 2.4. Assume that S is of characteristic zero at every point s such that we have $Y_s \neq \emptyset$. Then, if $\mathcal{F}$ is a locally constant constructible sheaf of groups on U (resp. a constructible stack on U which is, locally on X and S , the inverse image of a constructible stack on S), $(\mathcal{F}, i)$ is cohomologically proper relative to S in dimension ≤ 1 .*

Since every constructible sheaf of sets on U is tamely ramified over X relative to S , the corollary follows from 2.4 (resp. 2.5).

COROLLARY 2.7. *The notation is that of 2.4, but we are given in addition an S -scheme T , a proper morphism $p: X \rightarrow T$, and we assume X and T of finite presentation over S ; let $q = pi$. Let $\mathcal{F}$ be a constructible sheaf of sets on U satisfying one of the conditions a) or b) of 2.4 (resp. a sheaf of groups satisfying one of the conditions a), b) of 2.4, resp. a stack on U which is, locally on X and S , the inverse image of a constructible stack G on S). Then we have the following conclusions:*

- 1) $(\mathcal{F}, q)$ is cohomologically proper relative to S in dimension ≤ 0 (resp. for every morphism $h: S' \rightarrow S$, if $m = h_{(T)}$, the canonical morphism:

$$\Theta: m^*(R_t^1 q_* \mathcal{F}) \longrightarrow R_t^1 q'_* \mathcal{F}'$$

is an isomorphism, resp. for every morphism $S' \rightarrow S$, the canonical morphism

$$\xi: m^*(q_*^t \mathcal{F}) \longrightarrow q'^t_* \mathcal{F}'$$

is an equivalence).

- 2) the sheaf $q_* \mathcal{F}$ (resp. the sheaf $R_t^1 q_* \mathcal{F}$, resp. the stack $q_*^t \mathcal{F}$) is constructible. In the last case, if we assume S locally noetherian and G 1-constructible, the same is true of $q_*^t \mathcal{F}$.

The first part follows at once from 2.4, 2.5 and from the proof of 1.8. Let us prove 2). If $\mathcal{F}$ is a constructible sheaf of sets on U satisfying 2.4 a) or 2.4 b), it follows from 2.4 2) that $i_* \mathcal{F}$ is constructible; hence so is $q_* \mathcal{F} = p_*(i_* \mathcal{F})$ (SGA 4 XIV 1.1).

Let $\mathcal{F}$ be a constructible sheaf of groups on U satisfying 2.4 a) or 2.4 b) and let us prove that $R_t^1 q_* \mathcal{F}$ is constructible. By passage to the limit (EGA IV 8.10.5 and 17.7.8) and using 1), we may assume S noetherian. Let then Φ be the stack on X whose fiber at every scheme X' étale over X consists of the torsors on $U' = U \times_X X'$, with group $\mathcal{F}|U$, which are tamely ramified over X relative to S ; we therefore have

$$S(i_*^t \Phi) \simeq R_t^1 i_* \mathcal{F}$$

and this sheaf is constructible by 2.4 2). It therefore follows from 6.3 below that $S(p_* \Phi)$ is constructible, i.e. that $R_t^1 q_* \mathcal{F}$ is constructible.

Finally, if $\mathcal{F}$ is a stack on U which is, locally on X and S , the inverse image of a constructible stack on S , $i_*^t \mathcal{F}$ is constructible, and hence so is $q_*^t \mathcal{F} = p_*(i_*^t \mathcal{F})$. If moreover S is locally noetherian and SG constructible, $S(i_*^t \mathcal{F})$ is constructible by 6.3; hence so is $S(q_*^t \mathcal{F})$, i.e. $S(q_*^t \mathcal{F})$ 6.2.

COROLLARY 2.8. *Let*

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ g \downarrow & \swarrow f & \\ S & & \end{array}$$

be a commutative diagram of schemes, in which U is the open set complementary in X to a divisor with normal crossings relative to S , f a proper morphism of finite presentation. Let $\mathbb{L}$ be a set of prime numbers. We assume g locally 0-acyclic (resp. locally 1-aspherical for $\mathbb{L}$). Then, if $\mathcal{F}$ is a sheaf of sets on U (resp. a sheaf of $\mathbb{L}$ -groups) on U , locally constant constructible, tamely ramified over X relative to S , $f_* \mathcal{F}$ (resp. $R_t^1 f_* \mathcal{F}$) is locally constant constructible and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 0 (resp. the formation of $R_t^1 f_* \mathcal{F}$ commutes with every change of base $S' \rightarrow S$). In the case without “resp.”, if $\mathcal{F}$ is a sheaf of groups, for every specialization $\bar{s}_1 \rightarrow \bar{s}_2$ of geometric points of S , the specialization morphism

$$(R_t^1 f_* \mathcal{F})_{\bar{s}_2} \longrightarrow (R_t^1 f_* \mathcal{F})_{\bar{s}_1}$$

is injective.

The corollary follows at once from 2.6 and from 1.14 (resp. from the analogue of 1.14 for $R_t^1 f_* \mathcal{F}$, which is proved as in loc. cit.).

COROLLARY 2.9. *Let*

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ g \downarrow & \swarrow f & \\ S & & \end{array}$$

be a commutative diagram of schemes, in which U is the open set complementary in X to a divisor with normal crossings relative to S , f a proper smooth morphism of finite presentation. Let $\mathbb{L}$ be the set of prime numbers distinct from the residue characteristics of S . Let $\mathcal{F}$ be a locally constant constructible sheaf of $\mathbb{L}$ -groups on U , tamely ramified over X relative to S . Then $R^1 f_* \mathcal{F}$ is locally constant constructible and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 1 .

The corollary follows from 2.8 and from the fact that we have $R^1 f_* \mathcal{F} = R^1 f_* \mathcal{F}$ (2.3 b)).

2.10. If U is a connected scheme, a a geometric point of U , $\mathbb{L}$ a set of prime numbers, we denote by

$$(2.10.0) \quad \pi_1^{\mathbb{L}}(U, a)$$

the projective limit of the finite quotients of $\pi_1(U, a)$ whose orders have all their prime factors in $\mathbb{L}$.

We shall define specialization morphisms for the fundamental group, generalizing X.2.

Let $g: U \rightarrow S$ be a coherent morphism with geometrically connected fibers (resp. a morphism of the form $g = fi$, where $f: X \rightarrow S$ is a proper morphism of finite presentation and where $i: U \rightarrow X$ is an open immersion such that U is the complement in X of a divisor with normal crossings relative to S (cf. 2.8)). Let $\mathbb{L}$ be a set of prime numbers and suppose, in the non-resp. case, that, for every finite constant $\mathbb{L}$ -group C , (C_U, g) is cohomologically proper relative to S in dimension ≤ 1 . Let $\bar{s}_1 \rightarrow \bar{s}_2$ be a specialization morphism of geometric points of S , $\bar{S}$ the strict localization of S at $\bar{s}_2$, $\bar{U} = U \times_S \bar{S}$. We have a commutative diagram

$$\begin{array}{ccccc} U_{\bar{s}_1} & \xrightarrow{h_1} & \bar{U} & \xleftarrow{h_2} & U_{\bar{s}_1} \\ \downarrow & & \downarrow \bar{g} & & \downarrow \\ \bar{s}_1 & \longrightarrow & \bar{S} & \longleftarrow & \bar{s}_2. \end{array}$$

If a_1 is a geometric point of $U_{\bar{s}_1}$, a_2 a geometric point of $U_{\bar{s}_2}$, the morphisms h_1 and h_2 define canonical morphisms

$$\begin{aligned} \pi_1: \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) &\longrightarrow \pi_1^{\mathbb{L}}(\bar{U}, a_1) & \pi_2: \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) &\longrightarrow \pi_1^{\mathbb{L}}(\bar{U}, a_2) \\ (\text{resp. } \pi_1: \pi_1^t(U_{\bar{s}_1}, a_1) &\longrightarrow \pi_1^t(\bar{U}, a_1) & \pi_2: \pi_1^t(U_{\bar{s}_2}, a_2) &\longrightarrow \pi_1^t(\bar{U}, a_2)) \end{aligned}$$

(V 7 and 2.1.5.2). The hypotheses of cohomological properness (resp. 2.8) prove that π_2 is an isomorphism. If one chooses a class of paths from a_1 to a_2 , one obtains an isomorphism

$$\pi_{12}: \pi_1^{\mathbb{L}}(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(\bar{U}, a_2) \quad (\text{resp. } \pi_{12}: \pi_1^t(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^t(\bar{U}, a_2))$$

whence a morphism $\pi = \pi_2^{-1} \pi_{12} \pi_1$

$$\pi: \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) \longrightarrow \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) \quad (\text{resp. } \pi: \pi_1^t(U_{\bar{s}_1}, a_1) \longrightarrow \pi_1^t(U_{\bar{s}_2}, a_2)).$$

Changing the class of paths from a_1 to a_2 amounts to modifying π by an inner automorphism of $\pi_1^{\mathbb{L}}(X_{\bar{s}_2}, a_2)$ (resp. of $\pi_1^t(X_{\bar{s}_2}, a_2)$). We call *specialization morphism for the fundamental group* associated with the morphism $\bar{s}_1 \rightarrow \bar{s}_2$, and we denote simply by

$$\pi: \pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \longrightarrow \pi_1^{\mathbb{L}}(X_{\bar{s}_2}) \quad (\text{resp. } \pi: \pi_1^t(X_{\bar{s}_1}) \longrightarrow \pi_1^t(X_{\bar{s}_2}))$$

any one of the morphisms defined above.

LEMMA 2.11. Let $f: X \rightarrow S$ be a proper morphism of finite presentation, D a divisor on X with normal crossings relative to S , $Y = \text{Supp } D$, $U = X - Y$, $i: U \rightarrow X$ the canonical morphism, $\bar{s}_1 \rightarrow \bar{s}_2$ a specialization morphism of geometric points of S , y_1 a geometric point of $Y_{\bar{s}_1}$, y_2 a geometric point of $Y_{\bar{s}_2}$, such that the projection z_1 of y_1 on X is a generization of the projection z_2 of y_2 . Let I_{y_1} be an inertia subgroup of $\pi_1^t(\bar{U}_{\bar{s}_1})$ at y_1 . Then the image of I_{y_1} under the specialization morphism

$$\pi: \pi_1^t(U_{\bar{s}_1}) \longrightarrow \pi_1^t(U_{\bar{s}_2})$$

is an inertia subgroup of $\pi_1^t(U_{\bar{s}_2})$ at y_2 .

Indeed, let $\bar{X}$ (resp. $\tilde{X}$) be the strict localization of X at y_2 (resp. at y_1), $\bar{U} = U \times_X \bar{X}$ (resp. $\tilde{U} = U \times_X \tilde{X}$). We have a canonical morphism $\tilde{U} \rightarrow \bar{U}$, and it follows from 1.10 that we have a commutative diagram

$$\begin{array}{ccc} \pi_1^t(\tilde{U}_{\bar{s}_1}) & \xrightarrow{\pi'} & \pi_1^t(\bar{U}_{\bar{s}_2}) \\ \downarrow & & \downarrow \\ \pi_1^t(U_{\bar{s}_1}) & \xrightarrow{\pi} & \pi_1^t(U_{\bar{s}_2}) \end{array}$$

where π' is the composite of the canonical morphism $\pi_1^t(\tilde{U}_{\bar{s}_1}) \rightarrow \pi_1^t(\bar{U}_{\bar{s}_1})$ and of the specialization morphism. Since $\pi_1^t(\tilde{U}_{\bar{s}_1})$ (resp. $\pi_1^t(\bar{U}_{\bar{s}_1})$) is an inertia group of $\pi_1^t(U_{\bar{s}_1})$ at y_1 and (resp. of $\pi_1^t(U_{\bar{s}_2})$ at y_2), it suffices to prove that π' is surjective. But this follows from the expression obtained in 5.6.

COROLLARY 2.12. *Let X be a proper and smooth connected curve of genus g over a separably closed field k of characteristic $p \geq 0$. Let U be the open set obtained by removing from X n distinct closed points $a_1, \dots, a_n$. Then the tamely ramified fundamental group $\pi_1^t(U)$ (2.1.3) can be generated by $2g + n$ elements x_i, y_i, σ_j , with $1 \leq i \leq g$, $1 \leq j \leq n$, such that σ_j is a generator of an inertia group corresponding to a_j , and such that we have the relation*

$$(*) \quad \prod_{1 \leq i \leq g} (x_i y_i x_i^{-1} y_i^{-1}) \cdot \prod_{1 \leq j \leq n} \sigma_j = 1.$$

For every finite group G of order prime to p , generated by elements $\bar{x}_i, \bar{y}_i, \bar{\sigma}_j$ satisfying the relation $(*)$, there exists an étale covering of U , with group G , corresponding to a homomorphism $\pi_1^t(U) \rightarrow G$ which sends x_i, y_i, σ_j to $\bar{x}_i, \bar{y}_i, \bar{\sigma}_j$ respectively. In other words, if p' denotes the set of prime numbers distinct from p , $\pi_1^{p'}(U)$ is the pro- p' -group generated by the generators x_i, y_i, σ_j bound by the single relation $(*)$.

Proof. We may assume k algebraically closed. Suppose first k of characteristic zero. There then exists an algebraically closed subextension k' of k , of finite transcendence degree over $\mathbf{Q}$, such that X comes by extension of scalars from a proper and smooth curve X' defined over k' , and we may assume that the points $a_1, \dots, a_n$ come from rational points $a'_1, \dots, a'_n$ of X' . Since k' is of finite transcendence degree over $\mathbf{Q}$, we can find an embedding of k' into the field of complex numbers $\mathbf{C}$; let $\tilde{U} = U' \times_{k'} \mathbf{C}$. Let k'' be an algebraically closed extension of k' such that we have k' -morphisms of k and of $\mathbf{C}$ into k'' . If $g': U' \rightarrow k'$ is the structural morphism, and if $\mathcal{F}$ is a finite constant sheaf of groups on U'' , it follows from 2.9 that the specialization morphisms

$$(R^1 g'_* \mathcal{F}')_{\mathbf{C}} \longrightarrow (R^1 g'_* \mathcal{F}')_{k''} \longleftarrow (R^1 g'_* \mathcal{F}')_k$$

are isomorphisms. In terms of fundamental groups, this shows that we have an isomorphism, defined up to inner automorphism

$$\pi_1(U) \longrightarrow \pi_1(\tilde{U}),$$

and it is clear that this isomorphism transforms an inertia group relative to a point of $X' - U'$ into an inertia group relative to the same point. We may therefore assume that $k = \mathbf{C}$. In this last case it follows from Riemann's existence theorem (XII 5.2) that the fundamental group $\pi_1(U)$ is none other than the completion, for the topology of subgroups of finite index, of the fundamental group of the analytic space associated with U . Now the latter can be computed by transcendental means [3, ch. 7 §47]; it can be generated by $2g + n$ elements x_i, y_i, σ_j such that σ_j is the image of a generator of the local fundamental group $\pi_1(D_j)$ of a small disk centered at a_j , i.e. a generator of an inertia group corresponding to the point a_j , these elements satisfying the single relation $(*)$.

If now k is of characteristic $p > 0$, we can find a complete discrete valuation ring A , with residue field k , with field of fractions K of characteristic zero, and a connected scheme X_1 , proper and smooth over $S = \operatorname{Spec} A$, such that we have $X_1 \times_S \operatorname{Spec} k \simeq X$ (III 7.4). The points a_j then lift to sections s_j of X_1 over S ; let Y_{1j} be the reduced closed subscheme with underlying space $s_j(S)$, Y_1 the union of the Y_{1j} , $U_1 = X_1 - Y_1$, $g_1: U_1 \rightarrow S$ the structural morphism. Let $\overline{K}$ be an algebraically closed extension of K , $\overline{U} = U_1 \times_S \overline{K}$. If C is a finite constant group, it follows from 2.8 that the specialization morphism

$$(R^1 g_{1*} C_{U_1})_k \longrightarrow (R^1 g_{1*} C_{U_1})_{\overline{K}}$$

is injective and even bijective if C is of order prime to p . Now this means, in terms of fundamental groups, that the specialization morphism (1.10)

$$\pi: \pi_1(\overline{U}) \longrightarrow \pi_1^t(U)$$

is surjective, and that the specialization morphism

$$\pi_1^{p'}(\overline{U}) \longrightarrow \pi_1^{p'}(U)$$

is bijective. Finally, if x_i, y_i, σ_j are generators of $\pi_1(\overline{U})$ such that σ_j is a generator of an inertia group corresponding to the point $b_j = Y_{1j}(\overline{K})$ of $\overline{X}$, then, by 1.11, $\pi(\sigma_j)$ is a generator of an inertia group corresponding to a_j , which completes the proof.

[Remark 2.13 (added in 2003 (MR)). Let k be an algebraically closed field of characteristic $p > 0$, X a proper smooth algebraic curve over k , connected, of genus g , U an affine open set of X , the complement of $r \geq 1$ rational points of X . We have at our disposal the fundamental group $\pi_1(U)$, its quotient $\pi_1^t(U)$ which classifies the finite étale coverings of U tamely ramified at the points of $X - U$, and the quotient $\pi_1^{p'}(U)$ of $\pi_1^t(U)$ which classifies the Galois étale coverings of U with Galois group of order prime to p .

In (Coverings of algebraic curves, Amer. J. Math. **79** (1957), p. 825–856) S. Abhyankar formulated a certain number of conjectures on the structure of the finite groups that are Galois groups of finite étale coverings of U .

The conjectures concerning the finite groups that are Galois groups of connected étale coverings of U of order prime to p are proved and made more precise in Corollary 2.12. Before tackling the finite quotients of $\pi_1(U)$, let us begin by giving some indications about the “size” of this group. Let $X = \operatorname{Spec}(A)$. It is known that $\operatorname{Hom}_{\text{cont}}(\pi_1(U), \mathbf{Z}/p\mathbf{Z})$ is described by Artin-Schreier theory (cor. XI 6.9). This group is isomorphic to $A/\wp(A)$ where $\wp$ is the map from A to A which sends a to $a^p - a$. Suppose first that U is the affine line $\mathbf{A}^1$, with ring $k[T]$. Let E be the set of elements of $k[T]$ of the form $\sum_i a_i T^i$, with $a_i = 0$ when p divides i . It is immediate that the composite map $E \rightarrow k[T] \rightarrow k[T]/\wp k[T]$ is bijective. Consequently the coefficients a_i , $(i, p) = 1$, behave like coordinates of a space which parametrizes the cyclic coverings of degree p of the affine line. In particular, one deduces that $\pi_1(\mathbf{A}^1)$ is not topologically of finite type and that, if one makes an extension $k \rightarrow k'$ of algebraically closed fields, the natural morphism $\pi_1(\mathbf{A}^1 \times_k k') \rightarrow \pi_1(\mathbf{A}^1)$, which is surjective, is not bijective, contrary to the proper case. In the general case, the curve U is realized as a scheme finite over the affine line and one concludes that the same phenomena occur for $\pi_1(U)$ (and moreover, more generally, for $\pi_1(V)$ for every connected affine k -scheme V of finite type, of dimension ≥ 1).

This being so, if G is a finite group, let us denote by $G^{(p')}$ the largest quotient group of G of order prime to p . For a finite group G to be a topological quotient of $\pi_1(U)$, it is necessary that $G^{(p')}$ be a topological quotient of $\pi_1^{p'}(U)$, a condition which one knows in principle how to answer thanks to Corollary 2.12. In the aforementioned article, Abhyankar conjectures that this necessary condition is also sufficient. This conjecture was proved by M. Raynaud in the case of the affine line and by D. Harbater in the general case (Invent. Math. **116** (1994), p. 425–462 and **117**, p. 1–25). For example, in the case of the affine line $\pi_1^{p'}(\mathbf{A}^1) = 1$ and one concludes that a finite group G is the Galois group of a connected étale covering of $\mathbf{A}^1$ if and

only if $G^{(p')} = 1$, that is, if and only if G is generated by its Sylow p -subgroups. Thus every finite simple group of order a multiple of p will do.]

3. Cohomological properness and generic local acyclicity

THEOREM 3.1. *Let S be an irreducible scheme with generic point s , X and Y two S -schemes of finite presentation, $f: X \rightarrow Y$ an S -morphism. For every S -scheme S' , we denote by Y' , X' , etc. the inverse image of Y , X , etc. under the morphism $S' \rightarrow S$. We have the following properties:*

- 1) a) *One can find a nonempty open set S' of S such that, for every finite constant sheaf of sets F' on X' , f'_*F' is constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 0 .*
- b) *Let F be a constructible sheaf of sets on X . Then one can find a nonempty open set S' of S (depending on F) such that f'_*F' is constructible and (F', f') is cohomologically proper relative to S' in dimension ≤ 0 .*
- 2) *Suppose that the schemes of finite type of dimension $\leq \dim X_s$ over an algebraic closure $\bar{k}$ of $k(s)$ are strongly desingularizable (SGA 5 I 3.1.5). Then we have moreover the following properties:*
 - a) *One can find a nonempty open set S' of S such that, for every finite constant sheaf of groups F' on X' , of order prime to the residue characteristics of S , if Φ' is the stack of torsors under F' , $f'_*\Phi'$ is 1-constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 1 .*
 - b) *Let $\mathbb{L}$ be the set of prime numbers distinct from the residue characteristics of S and Φ a 1-constructible ind- $\mathbb{L}$ -stack on X (0), such that, for every scheme X_1 étale over X and for every pair of objects x, x_1 of Φ_{X_1} , the sheaf $\mathbf{Hom}_{X_1}(x, x_1)$ is constructible. Suppose moreover S locally noetherian. Then one can find a nonempty open set S' of S such that $f'_*\Phi'$ is 1-constructible, that, for every pair of objects y, y_1 of a fiber $(f'_*\Phi')_{Y_1}$, $\mathbf{Hom}_{Y_1}(y, y_1)$ is constructible, and that (Φ', f') is cohomologically proper relative to S' in dimension ≤ 1 .*

Proof. We may assume S affine; by SGA 4 VIII 1.1, we may assume S integral; finally, by passage to the limit, we may assume that S is the spectrum of an algebra of finite type over $\mathbf{Z}$; in particular S is then noetherian. Since the question is local on Y , we may assume Y affine. Moreover, to prove the theorem, it suffices to do so after a finite extension $S' \rightarrow S$, where S' is an integral scheme and where $S' \rightarrow S$ is a composite of étale morphisms and of finite radicial surjective morphisms.

1) Case of constant sheaves of sets

1) 1. Reduction to the case where X is normal over S . Let $X_{1\bar{s}}$ be the normalization of $(X_{\bar{s}})_{\text{red}}$; after restricting S to a nonempty open set and making a radicial extension of S if necessary, we may assume that $X_{1\bar{s}}$ comes from a scheme X_1 normal over S , and that the morphism $X_{1\bar{s}} \rightarrow X_{\bar{s}}$ comes from a finite surjective morphism $p: X_1 \rightarrow X$ (EGA IV 8.8.2 and 9.6.1). Suppose the theorem proved for fp . After restricting S to an open set if necessary, we may assume that, for every constant sheaf of sets F on X , (p^*F, fp) is cohomologically proper relative to S in dimension ≤ 0 and that $f_*p_*(p^*F)$ is constructible. By 1.9, (p_*p^*F, f) is then cohomologically proper relative to S in dimension ≤ 0 . The morphism

$$F \longrightarrow p_*p^*F = G$$

is a monomorphism. It already follows from this, f_*F being a subsheaf of f_*G , that f_*F is constructible (SGA 4 IX 2.9 (ii)) and that (F, f) is cohomologically proper relative to S in dimension ≤ -1 .

Let $X_2 = X_1 \times_X X_1$, $q: X_2 \rightarrow X$ the canonical morphism. By 1.11 1), we have an exact sequence

$$F \longrightarrow G \rightrightarrows q_*q^*F.$$

By what has just been proved, applied to fq instead of f , we may assume, after restricting S to a nonempty open set if necessary, that, for every constant sheaf of sets F on X , (q^*F, fq) is cohomologically proper relative to S in dimension ≤ -1 , hence that (q_*q^*F, f) is cohomologically proper relative to S in dimension ≤ -1 . It then follows from 1.13 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0 .

1) 2. Reduction to the case where X is normal affine over S .

Let U_s be an affine open set of X_s dense in X_s . After restricting S to a nonempty open set if necessary, we may assume that $U_s \rightarrow X_s$ lifts to an open immersion $i: U \rightarrow X$, schematically dominant relative to S (EGA IV 8.9.1). Since the morphism $X \rightarrow S$ is normal, we have by SGA 2 XIV 1.18:

$$\text{prof } \text{ét}_{S-U}(X) \geq 2;$$

consequently, for every constant sheaf F on X , the canonical morphism

$$F \longrightarrow i_*i^*F$$

is an isomorphism. It follows that, if we suppose the theorem proved for fi and i , then, after restriction of S to a nonempty open set, (i^*F, i) and (i^*F, fi) are cohomologically proper relative to S in dimension ≤ 0 . The same therefore holds for (F, f) (1.6 2)). Since moreover $f_*F = (fi)_*(i^*F)$ is constructible, this completes the reduction.

1) 3. End of the proof.

We may assume S normal (EGA IV 7.8.3). We can find a compactification of X_s :

$$\begin{array}{ccc} X_s & \xrightarrow{j_s} & P_s \\ f_s \downarrow & \swarrow g_s & \\ Y_s & & \end{array}$$

where j_s is a dominant open immersion and g_s a proper morphism; after making a radicial extension of $k(s)$ and replacing P_s by its normalization, which does not change X_s , we may assume P_s geometrically normal. After restricting S to a nonempty open set and making a surjective radicial extension if necessary, we may assume that the diagram above comes from a diagram

$$\begin{array}{ccc} X & \xrightarrow{j} & P \\ f \downarrow & \swarrow g & \\ Y & & \end{array}$$

where P is a scheme normal over S , j an open immersion schematically dominant relative to S and g a proper morphism (EGA IV 6.9.1, 9.9.4 and 9.6.1). For every finite constant sheaf of sets F on X with value C , j_*F is the constant sheaf with value C (SGA 4 2.14.1), and the same holds after any change of base $S' \rightarrow S$; it follows that (F, j) is cohomologically proper relative to S in dimension ≤ 0 . The same therefore holds for (F, f) , since g is proper (1.8). Since g is proper $f_*F = g_*C_P$ is constructible, which completes the proof of 1) a).

2) Case of a constructible sheaf of sets

Let F be a constructible sheaf of sets on X . By SGA 4 IX 2.14 (ii), we can find a finite family of morphisms $p_i: Z_i \rightarrow X$, and, on each Z_i , a finite constant sheaf of sets C_i , such that we have a monomorphism

$$j: F \longrightarrow \prod_i p_{i*}C_i = G.$$

By 1) a), we may assume, after restricting S to a nonempty open set if necessary, that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 0 , and that the $f_*p_{i*}C_i$ are constructible. We already conclude from this that (G, f) is cohomologically proper relative to S in dimension ≤ 0 (1.9), hence that (F, f) is cohomologically proper relative

to S in dimension ≤ -1 , and that f_*F is constructible. Let K be the amalgamated sum $K = G \amalg_F G$; since F and G are constructible, so is K . We therefore conclude from the preceding that, after restricting S to a nonempty open set if necessary, we may assume that (K, f) is cohomologically proper relative to S in dimension ≤ -1 . It then follows from 1.13 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0 .

3) Case of constant sheaves of groups

If F is a constant sheaf of groups on X , we denote by Φ the stack of torsors under F .

3) 1. Let us first show that, after restricting S to a nonempty open set if necessary, for every constant sheaf of groups F on X , of order prime to the residue characteristics of S , (F, f) is cohomologically proper relative to S in dimension ≤ 0 , and that $f_*\Phi$ is constructible.

For this we reduce to the case where X is smooth over S . After making a finite extension of $k(s)$ if necessary, which is permissible since one can regard it as the composite of an étale extension and of a radicial extension, we can find a proper surjective morphism $p_s: X_{1s} \rightarrow X_s$, where X_{1s} is a scheme smooth over S of the same dimension as X_s , and, after restricting S to a nonempty open set if necessary, we may assume that p_s comes from a proper surjective morphism $p: X_1 \rightarrow X$, where X_1 is a scheme smooth over S (EGA IV 9.6.1 and 12.1.6). Let $X_2 = X_1 \times_X X_1$, $q: X_2 \rightarrow X$ the canonical morphism. We have an exact diagram of stacks on X

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi$$

(1.11 2)). The theorem being assumed proved in the smooth case, we see first of all that we may assume $f_*p_*p^*\Phi$ constructible; hence so is $f_*\Phi$ (3.1.1 below). Moreover, by 1.6 2), we may assume $(p_*p^*\Phi, f)$ cohomologically proper relative to S in dimension ≤ 0 ; it follows that (Φ, f) is cohomologically proper relative to S in dimension ≤ -1 . We may therefore assume that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1 , and this implies that (Φ, f) is cohomologically proper relative to S in dimension ≤ 0 (1.12 1)).

We then reduce as in 1) 2 to the case where X is smooth and affine over S . Let then

$$\begin{array}{ccc} X & \xrightarrow{i} & P \\ f \downarrow & \swarrow q & \\ Y & & \end{array}$$

be a compactification of X , where i is a dominant open immersion and q a proper morphism. Since we have $\dim P_s = \dim X_s$, we can apply the hypothesis of resolution of singularities to P_s . After making an étale extension and a radicial extension of S if necessary, we can find a proper morphism $r: Z \rightarrow P$, where Z is smooth over S , $r^{-1}(X) \simeq X$, and where $r^{-1}(X)$ is the complement in Z of a divisor with normal crossings relative to S . Every torsor under F is then tamely ramified on Z relative to S (2.3 b)). It therefore follows from 2.7 that (F, f) is cohomologically proper relative to S in dimension ≤ 0 , which proves our assertion.

3) 2. Reduction to the case where X is smooth over S .

After making a finite extension of $k(s)$ if necessary, we can find a proper surjective morphism $p_s: X_{1s} \rightarrow X_s$, where X_{1s} is a scheme smooth over s , and we may assume that p_s comes from a proper surjective morphism $p: X_1 \rightarrow X$, where X_1 is smooth over S . Suppose the theorem proved for fp and let us prove it for f . Let F be a finite constant sheaf of groups on X , of order prime to the residue characteristics of S and Φ the stack of torsors under F . Let $X_2 = X_1 \times_X X_1$, $X_3 = X_1 \times_X X_1 \times_X X_1$, $q: X_2 \rightarrow X$, $r: X_3 \rightarrow X$ the canonical morphisms. By 1.11 2), we have an exact diagram of stacks

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi \rightrightarrows r_*r^*\Phi.$$

To prove that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 , it suffices to show that the same holds for $(p_*p^*\Phi, f)$, that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0 and that $(r_*r^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1 (1.12 2)). By 3) 1 above we may assume that, for every finite constant sheaf of groups F ,

$(q^*\Phi, fq)$, $(q^*\Phi, q)$, $(r^*\Phi, fr)$, $(r^*\Phi, r)$ are cohomologically proper relative to S in dimension ≤ 0 . It then follows from 1.6 2) that $(q_*q^*\Phi, f)$ and $(r_*r^*\Phi, f)$ are cohomologically proper relative to S in dimension ≤ 0 . The theorem being assumed proved in the smooth case, $(p^*\Phi, fp)$ and $(p^*\Phi, p)$, hence also $(p_*p^*\Phi, f)$ (1.6 2)), are cohomologically proper relative to S in dimension ≤ 1 . This indeed shows that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 .

Moreover $f_*p_*p^*\Phi$ is 1-constructible by hypothesis; by 3.1 we may assume that $f_*q_*q^*\Phi$ is constructible; it therefore follows from 3.1.1 below that $f_*\Phi$ is 1-constructible.

3) 3. Reduction to the case where X is smooth affine over S .

By 3) 2), we may assume X smooth over S . Let $(U_i)_{i \in I}$ be a finite covering of X by affine open sets, and let X_1 be the direct sum of the U_i , $p: X_1 \rightarrow X$ the canonical morphism. Since p is an effective descent morphism for the category of étale sheaves of finite type over variable schemes, we see as in 3) 2) that, if we suppose the theorem proved for the U_i , i.e. for X_1 , it is also true for X .

3) 4. Case where X is smooth affine over S .

We see as in 3)1 that, after restricting S to a nonempty open set and making a surjective étale extension and a surjective radicial extension of S if necessary, we can find a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & P \\ f \downarrow & \swarrow q & \\ Y & & \end{array}$$

where P is a scheme smooth over S , X the complement in P of a divisor with normal crossings relative to S and g a proper morphism. If F is a constant sheaf of order prime to the residue characteristics of S , every torsor under F is tamely ramified on P relative to S (2.3 b)). The fact that (F, f) is cohomologically proper relative to S in dimension ≤ 1 and that $f_*\Phi$ is constructible then follows from 2.7.

4) Proof of 2) b)

4) 1. Case where Φ is a gerbe.

One can find a surjective étale morphism of finite type $p: X_1 \rightarrow X$ such that $p^*\Phi$ is a trivial gerbe. By descent, as in 3) 2), one sees that it suffices to prove the theorem for X_1 , $X_1 \times_X X_1$ and $X_1 \times_X X_1 \times_X X_1$. One is therefore reduced to the case where Φ is the gerbe of torsors under a constructible sheaf of groups F whose fibers are of order prime to the residue characteristics of S .

By SGA 4 IX 2.14, one can find a finite family of finite morphisms $p_i: Z_i \rightarrow X$ and, for each i , a finite constant sheaf of groups C_i , of order prime to the residue characteristics of S , such that one has a monomorphism

$$j: F \longrightarrow \prod_i p_{i*}C_i = G.$$

Let Φ_i be the stack of torsors under C_i and Ψ the stack of torsors under G . It follows from 2) a) that, at the cost of restricting S to a nonempty open subset, one may assume that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 1 , and that the stacks $f_*p_{i*}\Phi_i$ are 1-constructible. It then follows from 1.9 that the $(p_{i*}C_i, f)$ are cohomologically proper relative to S in dimension ≤ 1 ; hence the same holds for (G, f) . Moreover, since $p_{i*}\Phi_i$ is equivalent to the stack of torsors under the group $p_{i*}C_i$ (SGA 4 VIII 5.8), one sees that $f_*\Psi$ is 1-constructible.

Since R^1f_*G is constructible, one can find a sheaf representable by an étale Y -scheme of finite type T and an epimorphism

$$a: T \longrightarrow R^1f_*G$$

(SGA 4 IX 2.7); moreover one may assume that the image of the identity section of $T(T)$ is defined by a torsor Q on $X \times_Y T = X_T$, with group $G|X_T$. Let $f_T: X_T \rightarrow T$ be the canonical morphism, $F_T = F|X_T$, etc. By 1) b) one may assume, at the cost of restricting S to a nonempty open subset, that $(Q/F_T, f_T)$ is cohomologically proper relative to S in dimension ≤ 0 and that $f_{T*}(Q/F_T)$ is constructible. It then follows from 3.1.2 that $f_*\Psi$ is constructible.

Let us show that (F, f) is cohomologically proper relative to S in dimension ≤ 1 . By 1.13 2) it suffices to prove that, for every scheme Y_1 étale over Y and every torsor Q_1 on $X_1 = X \times_Y Y_1$, if $f_1: X_1 \rightarrow Y_1$ is the canonical morphism, then $(Q_1/F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0 . Now, by definition of T , Q_1 is, locally for the étale topology of Y_1 , an inverse image of Q , which proves our reduction.

4) 2. General case.

Using lemma 6.1.1, 4) 1 and 1) a), one sees that, at the cost of restricting S to a nonempty open subset, one may assume $S(f_*\Phi)$ constructible and $(S\Phi, f)$ cohomologically proper relative to S in dimension ≤ 0 . One can then find a sheaf representable by an étale Y -scheme of finite type T and an epimorphism

$$a: T \longrightarrow S(f_*\Phi).$$

One takes up again the notation of 4) 1, and one sets moreover $Z = T \times_Y T$, $X_Z = X \times_Y Z$, and one denotes by $f_Z: X_Z \rightarrow Z$ the canonical morphism. One may assume T chosen so that the image q of the identity section of $T(T)$ under a is defined by an object p of $(f_*\Phi)_T = \Phi_{X_T}$. Let p_1 and p_2 (resp. q_1 and q_2) be the inverse images of p (resp. q) under the two projections of Z into T . One may assume, at the cost of restricting S to a nonempty open subset, that $(\mathbf{Aut}_{X_T}(p), f_T)$ is cohomologically proper relative to S in dimension ≤ 1 , that $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0 , and that $f_{T*}(\mathbf{Aut}_{X_T}(p))$, $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$ are constructible (a) 1) and 4) 1)).

From this one first deduces that, for every scheme Y_1 étale over Y and every pair of objects y, y_1 of $(f_*\Phi)_{Y_1}$, the sheaf $\mathbf{Aut}_{Y_1}(y)$ (resp. $\mathbf{Hom}_{Y_1}(y, y_1)$) is constructible, such a sheaf being, locally for the étale topology of Y_1 , an inverse image of $f_{T*}(\mathbf{Aut}_{X_T}(p))$ (resp. of $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$).

It remains to prove that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1 . For this it suffices to show that, for every S' -scheme S and every geometric point $\bar{y}'$ of Y , if one denotes by $\bar{Y}$ the strict localization of Y at y' , $\bar{X} = X \times_Y \bar{Y}$, etc., then the canonical functor

$$\bar{\varphi}: \Phi(\bar{X}) \longrightarrow \Phi'(\bar{X}')$$

is an equivalence of categories.

Let us show that $\bar{\varphi}$ is fully faithful. Let $x, y \in \Phi(\bar{X})$, let x', y' be their images in $\Phi'(\bar{X}')$, and let us show that the canonical morphism

$$(*) \quad \mathrm{Hom}_{\bar{X}}(x, y) \longrightarrow \mathrm{Hom}_{\bar{X}'}(x', y')$$

is bijective. By definition of T there exist two morphisms from $\bar{Y}$ to T such that x and y are the inverse images of p under these two morphisms. This amounts to saying that there is a morphism $\bar{Y} \rightarrow Z$, whence a morphism $h: \bar{X} \rightarrow X_Z$ such that one has

$$h^*(p_1) = x \quad h^*(p_2) = y.$$

Consequently one has a canonical isomorphism

$$\mathbf{Hom}_{\bar{X}}(x, y) = h^*(\mathbf{Hom}_{X_Z}(p_1, p_2)).$$

but, taking into account the fact that $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0 , one sees that the same holds for $(\mathbf{Hom}_{\bar{X}}(x, y), f)$, which proves that the morphism $(*)$ is bijective.

Let us show that $\bar{\varphi}$ is essentially surjective. Let $x' \in \Phi'(\bar{X}')$. Since $(S\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0 , the canonical morphism

$$H^0(\bar{X}, \bar{S}\Phi) \longrightarrow H^0(\bar{X}', \bar{S}\Phi')$$

is bijective. Let G' be the maximal subgerbe of Φ' generated by x' ; there then exists a maximal subgerbe G of $\bar{\Phi}$ whose inverse image on $\bar{X}'$ is G' . By 4) 1, $(G, \bar{f})$ is cohomologically proper relative to S in dimension ≤ 1 ; consequently the canonical functor

$$G(\bar{X}) \longrightarrow G'(\bar{X}')$$

is an equivalence of categories, which proves the existence of an element x of $\Phi(\bar{X})$ whose image in $\Phi'(X')$ is isomorphic to x' , and completes the proof of the theorem.

LEMMA 3.1.1. *Let S be a locally noetherian scheme and*

$$\Phi \xrightarrow{p} \Phi_1 \xrightarrow{p_1, p_2} \Phi_2$$

an exact diagram of stacks on S (1.10.1). If Φ_1 is constructible, so is Φ . If, for every scheme S' étale over S and every pair of objects x_1, y_1 of $(\Phi_1)_{S'}$, the sheaf $\mathbf{Hom}_{S'}(x_1, y_1)$ is constructible, then, for every pair of objects x, y of $\Phi_{S'}$, the same holds for $\mathbf{Hom}_{S'}(x, y)$. Suppose that Φ_1 is 1-constructible (0) and that Φ_2 is constructible; then Φ is 1-constructible.

For every scheme S' étale over S and every object x of $\Phi_{S'}$, one has a monomorphism

$$\mathbf{Aut}_{S'}(x) \longrightarrow \mathbf{Aut}_{S'}(p(x)).$$

It therefore follows from SGA 4 IX 2.9 that, if Φ_1 is constructible, so is Φ , and the second assertion of the lemma is proved in the same way.

Suppose now Φ_1 1-constructible and Φ_2 constructible. The morphism p induces on the sheaves of maximal subgerbes a morphism

$$\varphi: S\Phi \longrightarrow S\Phi_1.$$

Let G be the image of $S\Phi$ under φ ; by SGA 4 IX 2.9, G is a constructible sheaf. One can therefore find a sheaf representable by an étale S -scheme of finite type T and an epimorphism

$$a: T \longrightarrow G$$

(SGA 4 IX 2.7). Moreover one can choose T so that the image y of the identity section of $T(T)$ under a is defined by an object x_1 of $(\Phi_1)_T$ of the form $x_1 = p(x)$, where $x \in \Phi_T$.

It suffices to show that, for every point s of S , there exists a nonempty open subset U of $\{\bar{s}\}$ such that $S\Phi|U$ is locally constant constructible. Let $s \in S$, let $\bar{s}$ be a geometric point over s and $\bar{y}_1, \dots, \bar{y}_n$ the elements of $G_{\bar{s}}$. By definition of T there exist morphisms $h_i: \bar{s} \rightarrow T$ such that one has $h_i^*(y) = \bar{y}_i$. Let S' be the fiber product over S of n schemes isomorphic to T , $h: \bar{s} \rightarrow S'$ the fiber product of the h_i , y_i (resp. x_i) the inverse image of y (resp. x) under the i th projection of S' onto T . Let F_i be the subsheaf of $S\Phi|S'$ that is the inverse image of y_i , and let us show that the F_i are constructible. The sheaf F_i is a quotient of the sheaf F'_i such that, for every scheme S'' étale over S' , one has

$$F'_i(S'') = \{\text{classes mod. isomorphism of objects } z \text{ of } \Phi_{S''} \text{ equipped} \\ \text{with an isomorphism } i: p(z) \simeq p(x_i|S'')\}.$$

It suffices to show that the F'_i are constructible. Now, if one sets $z_i = p_1 p(x_i)$, $z'_i = p_2 p(x_i)$, one has a monomorphism

$$\Psi_i: F'_i \longrightarrow \mathbf{Isom}_{S'}(z_i, z'_i),$$

obtained by associating, with every scheme S'' étale over S' and every object z of $\Phi_{S''}$ such that one has an isomorphism $i: p(z) \simeq p(x_i|S'')$, the isomorphism from z_i to z'_i defined by the condition that the diagram

$$\begin{array}{ccc} p_1 p(z) & \xrightarrow{p_1(i)} & z_i \\ j \downarrow & & \downarrow \\ p_2 p(z) & \xrightarrow{p_2(i)} & z'_i \end{array}$$

be commutative (j is the canonical morphism associated with the exact diagram).

The morphism Ψ_i is injective because to say that two objects z, z' such that one has isomorphisms $i: p(z) \xrightarrow{\sim} p(x_i|S'')$, $i': p(z') \xrightarrow{\sim} p(x_i|S'')$ define the same element of $\mathbf{Isom}_{S''}(z_i, z'_i)$ amounts to saying that one has $p_1(i'^{-1}i) = p_2(i'^{-1}i)$, i.e. that $i'^{-1}i$ comes from an isomorphism $z \rightarrow z'$. The sheaf F'_i , being a subsheaf of $\mathbf{Isom}_{S'}(z_i, z'_i)$, is constructible.

Since one can find a nonempty open subset U of $\{s\}$ such that $y_1|U, \dots, y_n|U$ generate G , it follows from lemma 6.1.2 below that $S\Phi|U$ is constructible, hence, at the cost of shrinking U , $S\Phi|U$ is locally constant constructible.

LEMMA 3.1.2. *Let S be a locally noetherian scheme, $f: X \rightarrow S$ a morphism, $F \rightarrow G$ a monomorphism of sheaves of groups on X , Ψ (resp. Ψ_1) the stack of torsors under F (resp. under G), $\Phi = f_*\Psi$, $\Phi_1 = f_*\Psi_1$. Suppose given a sheaf on S representable by an étale S -scheme of finite type T and a surjective morphism*

$$a: T \longrightarrow S\Phi_1 \simeq R^1 f_* G,$$

such that there exists a torsor Q on $X_T = X \times_S T$ with group $G|_{X_T}$ which defines in $R^1 f_ G(T)$ the image under a of the identity section of $T(T)$. Let $f_T: X_T \rightarrow T$ be the canonical morphism and set $F_T = F|_{X_T}$. Suppose that Φ_1 is 1-constructible and that $f_{T*}(Q/F_T)$ is constructible. Then Φ is 1-constructible.*

It suffices indeed to copy the proof of 3.1.1, the fact that one has morphisms Ψ_i being replaced by the fact that one has isomorphisms

$$F'_i \xrightarrow{\sim} f_*(Q/F_T)|_{S'}.$$

REMARK 3.1.3. Suppose that $k = k(s)$ is of characteristic zero; then the schemes of finite type over $\bar{k}$ of dimension $\leq \dim X$ are strongly desingularizable, and the proof of 3.1 makes it possible to prove the following results:

- a) There exists a nonempty open subset S_1 of S such that, for every scheme S' over S_1 whose maximal points are of characteristic zero and every locally constant constructible sheaf of sets F on $X' = X \times_S S'$, (F, f') is cohomologically proper relative to S' in dimension ≤ 0 .
- b) If all the residue characteristics of S are zero, there exists a nonempty open subset S_1 of S such that, for every scheme S' over S_1 and every locally constant constructible sheaf of groups F on X' , (F, f') is cohomologically proper relative to S' in dimension ≤ 1 .

It suffices indeed to copy the proof of 3.1.2) a). Proposition 2.7 used in 3) 4 applies to the case of a locally constant sheaf F , because, taking up again the notation of 3) 4, every torsor under F is tamely ramified on P relative to S , since all the residue characteristics of S are zero.

COROLLARY 3.2. *Let k be a field of characteristic $p \geq 0$, p' the set of prime numbers distinct from p , $f: X \rightarrow k$ a coherent morphism.*

- 1) *For every sheaf of sets F , (F, f) is cohomologically proper in dimension ≤ 0 .*
- 2) *Suppose that one of the following two conditions is satisfied:*
 - a) *f is of finite type and the schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k are strongly desingularizable.*
 - b) *The schemes of finite type over an algebraic closure of k are strongly desingularizable.*

Then, for every sheaf of ind- p' -groups F , (F, f) is cohomologically proper in dimension ≤ 1 .

Let F be a sheaf of sets (resp. of ind- p' -groups). By SGA 4 IX 2.7.2 one can write F as a filtered inductive limit

$$F = \varinjlim F_i,$$

where the F_i are constructible sheaves of sets (resp. of ind- p' -groups). Since f is coherent, f_* (resp. R^1f_*) commutes with inductive limits (SGA 4 VII 3.3). If one knows that the (F_i, f) are cohomologically proper in dimension ≤ 0 (resp. that every sheaf of sets is cohomologically proper in dimension ≤ 0 and that the (F_i, f) are cohomologically proper in dimension ≤ 1), the same will hold for (F, f) . One may therefore assume F constructible.

If one assumes f of finite type (resp. satisfying a)), the proposition follows from 3.1 1) b) (resp. from 3.1 2) b)). Let us now prove 3.2 when one no longer assumes f of finite type. For every scheme S' over k and every geometric point $\bar{s}$ of S' , one denotes by $\bar{k}$ (resp. $\bar{S}'$) the strict localization of k at $\bar{s}$ (resp. of S' at $\bar{s}$), by $\bar{X}$ the inverse image of X over $\bar{k}$, and one considers the cartesian square

$$\begin{array}{ccc} \bar{X} & \xleftarrow{g} & \bar{X}' \\ \bar{f} \downarrow & & \downarrow \bar{f}' \\ \bar{k} & \xleftarrow{\quad} & \bar{S}' \end{array}$$

It suffices to prove that, for every S' and every $\bar{s}$, the canonical morphism

$$H^0(\bar{X}, \bar{F}) \longrightarrow H^0(\bar{X}', \bar{F}') \quad (\text{resp. } H^1(\bar{X}, \bar{F}) \longrightarrow H^1(\bar{X}', \bar{F}'))$$

is an isomorphism. It suffices to show that one has the relations

$$(*) \quad \bar{F} \simeq g_* g^* \bar{F} \quad (\text{resp. } R^1 g_*(g^* \bar{F}) = 0).$$

Now, in this form, the question is local on X for the étale topology. One may therefore assume X affine; by passage to the limit one may assume X of finite type over k . One then knows that (F, f) is cohomologically proper in dimension ≤ 0 (resp. ≤ 1) and that the same holds when X is replaced by a scheme étale of finite type over X , which proves (*).

THEOREM 3.3. *Let S be an irreducible scheme with generic point s , $f: X \rightarrow S$ a morphism of finite presentation. Suppose that the schemes of finite type of dimension $\leq \dim X_s$ over an algebraic closure $\bar{k}$ of k are desingularizable (EGA IV 7.9.1). Then, if $\mathbb{L}$ denotes the set of prime numbers distinct from the residue characteristics of S , one can find a nonempty open subset S_1 of S such that the morphism $f|_{S_1}$ is universally locally 1-aspherical for $\mathbb{L}$.*

One may assume S integral and X reduced (SGA 4 VIII 1.1). By passage to the limit one may assume S noetherian. Moreover, to prove the theorem, it suffices to do so after a finite extension $S_1 \rightarrow S$, where S_1 is an integral scheme and where $S_1 \rightarrow S$ is a composite of étale extensions and of surjective radicial extensions.

Let us first show that, at the cost of restricting S to a nonempty open subset, f is universally locally 0-acyclic. At the cost of making a radicial extension of $k(s)$, one may assume that the morphism $(X_s)_{\text{red}} \rightarrow s$ is separable; one may therefore assume, at the cost of restricting S to a nonempty open subset and of making a surjective radicial extension of S , that the morphism f is flat, with separable geometric fibers (EGA IV 12.1.1), which implies that f is universally 0-acyclic (SGA 4 XV 4.1).

Let us show that, at the cost of restricting S to a nonempty open subset, f is universally locally 1-aspherical for $\mathbb{L}$. At the cost of making a finite extension of $k(s)$, which is permissible because one can consider it as a composite of an étale extension and of a radicial extension, one can find a surjective proper morphism $p_s: Y_s \rightarrow X_s$, where Y_s is a scheme smooth over S , of the same dimension as X_s , and one may assume that p_s comes from a surjective proper morphism $p: Y \rightarrow X$, where Y is a scheme smooth over S (EGA IV 9.6.1 and 12.1.6). It suffices to show that, at the cost of restricting S to a nonempty open subset, for every diagram

with cartesian squares

$$\begin{array}{ccc} S'' & \xleftarrow{f''} & X'' \\ i \downarrow & & \downarrow j \\ S' & \xleftarrow{f'} & X' \\ \downarrow & & \downarrow \\ S & \xleftarrow{f} & X, \end{array}$$

where i is étale of finite presentation, and for every sheaf of ind- $\mathbb{L}$ -groups F on S'' , if Φ is the stack of torsors under F , then the canonical morphism

$$f'^* i_* \Phi \longrightarrow j_* f''^* \Phi$$

is an equivalence. Let $Z = Y \times_X Y, T = Y \times_X Y \times_X Y$. One has in a natural way a commutative diagram

$$\begin{array}{ccccccc} S'' & \xleftarrow{f''} & X'' & \xleftarrow{p''} & Y'' & \xleftarrow{\quad} & Z'' \xleftarrow{\quad} T'' \\ i \downarrow & & j \downarrow & & \downarrow & & \downarrow \\ S' & \xleftarrow{f'} & X' & \xleftarrow{p'} & Y' & \xleftarrow{\quad} & Z' \xleftarrow{\quad} T' \\ \downarrow & & \downarrow & & \downarrow & & \downarrow \\ S & \xleftarrow{f} & X & \xleftarrow{p} & Y & \xleftarrow{\quad} & Z' \xleftarrow{\quad} T' \end{array}$$

Let $q: Z \rightarrow X$ and $r: T \rightarrow X$ be the canonical morphisms, the notations q', r', q'', r'' having an obvious meaning. By 1.11 2), one has the following essentially commutative diagram, whose rows are exact:

$$\begin{array}{ccccccc} f'^* i_* \Phi & \longrightarrow & p'_* p'^* (f'^* i_* \Phi) & \Longrightarrow & q'_* q'^* (f'^* i_* \Phi) & \Longrightarrow & r'_* r'^* (f'^* i_* \Phi) \\ a \downarrow & & b \downarrow & & c \downarrow & & d \downarrow \\ j_* f''^* \Phi & \longrightarrow & j_* p''_* p''^* (f''^* \Phi) & \Longrightarrow & j_* q''_* q''^* (f''^* \Phi) & \Longrightarrow & j_* r''_* r''^* (f''^* \Phi) \end{array}$$

Since Y is smooth over S , the morphism fp is universally locally 1-aspherical for $\mathbb{L}$ (SGA XV 2.1), and it follows from [2, VII 2.1.7] that b is an equivalence of categories. On the other hand, at the cost of restricting S to a nonempty open subset, one may assume that the morphisms $Z \rightarrow S$ and $T \rightarrow S$ are universally locally 0-acyclic. It follows that the functors c and d are fully faithful, and the diagram above then shows that a is an equivalence, which completes the proof.

COROLLARY 3.4. *Let k be a field of characteristic $p \geq 0$, p' the set of prime numbers distinct from p , $f: X \rightarrow k$ a coherent morphism. Suppose that one of the following two conditions is satisfied:*

- a) *f is of finite type and the schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k are desingularizable.*
- b) *The schemes of finite type over an algebraic closure of k are desingularizable.*

Then f is universally locally 1-aspherical for p' .

Case a) follows from 3.3. In case b), the question being local on X , one may assume X affine; by passage to the limit (SGA 4 XV 1.3), one reduces to the case where X is of finite type over k .

COROLLARY 3.5. *Let S be an irreducible scheme with generic point s , $f: X \rightarrow S$ a morphism of finite presentation. Suppose that the schemes of finite type of dimension $\leq \dim X_s$ over an algebraic closure $\bar{k}$ of $k(s)$ are strongly desingularizable (SGA 5 I 3.1.5). If*

$\mathbb{L}$ denotes the set of prime numbers distinct from the residue characteristics of S , one can find a nonempty open subset S_1 of S such that, for every specialization $\bar{s}_1 \rightarrow \bar{s}_2$ of geometric points of S_1 , the specialization morphism (2.10)

$$\pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \longrightarrow \pi_1^{\mathbb{L}}(X_{\bar{s}_2})$$

is bijective.

By 3.1 and 3.3, one may, at the cost of restricting S to a nonempty open subset, assume that f is locally 1-aspherical for $\mathbb{L}$, and that, for every finite constant sheaf of $\mathbb{L}$ -groups F on X , (F, f) is cohomologically proper in dimension ≤ 1 . It then follows from 1.14 that, for every specialization $\bar{s}_1 \rightarrow \bar{s}_2$ of geometric points of S_1 , the specialization morphism

$$(R^1 f_* F)_{\bar{s}_2} \longrightarrow (R^1 f_* F)_{\bar{s}_1}$$

is bijective. The corollary is nothing but the translation of the foregoing in terms of fundamental groups.

4. Homotopy exact sequences

4.0. Let X and S be two connected schemes, $f: X \rightarrow S$ a morphism, a a geometric point of X , $\mathbb{L}$ a set of prime numbers. Let K be the kernel of the canonical homomorphism $\pi_1(X, a) \rightarrow \pi_1(S, a)$ and N the smallest normal pro-subgroup of K such that K/N is a pro- $\mathbb{L}$ -group $K^{\mathbb{L}}$. Then N is normal in $\pi_1(X, a)$ and we denote by

$$\pi'_1(X, a)$$

the quotient of $\pi_1(X, a)$ by N . If a is a geometric point of a geometric fiber $X_{\bar{s}}$, the canonical morphisms

$$\pi_1(X_{\bar{s}}, a) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(S, a)$$

allow one to define canonical morphisms

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a)$$

We have $vu = 0$.

PROPOSITION 4.1. *Let S be a connected scheme, $f: X \rightarrow S$ a locally 0-acyclic morphism (SGA 4 XV 1.11); suppose moreover that f is 0-acyclic (which, when f is coherent, amounts to saying that the geometric fibers of f are connected (SGA 4 XV 1.16)). Let $\mathbb{L}$ be a set of prime numbers. If S' is a scheme étale over S , we denote by X' , f' the inverse images of X , f over S' . Suppose that, for every étale covering S' of S and for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$ -group, (E, f') is cohomologically proper relative to S' in dimension ≤ 0 and $f'_* E$ is constructible. Then, if $\bar{s}$ is a geometric point of S and a a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$(4.1.1) \quad \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a) \longrightarrow 1$$

is exact

This statement generalizes X.1.4, whose proof we shall copy.

Let us first show that v is surjective. It suffices to show that, for every connected étale covering S' of S , X' is also connected (V 6.9). Let C be a set having at least two elements. It follows from the fact that f is 0-acyclic that the canonical morphism

$$H^0(S', C_{S'}) \longrightarrow H^0(X', C_{X'})$$

is bijective, hence that X' is connected, whence the surjectivity of v .

By definition of $K^{\mathbb{L}}$ (4.0), we have the exact sequence

$$1 \longrightarrow K^{\mathbb{L}} \longrightarrow \pi'_1(X, a) \longrightarrow \pi_1(S, a) \longrightarrow 1$$

Let $\tilde{S}$ be the universal covering of S and $\tilde{X} = \tilde{S} \times_S X$; the group $K^{\mathbb{L}}$ classifies the Galois coverings P whose group is an $\mathbb{L}$ -group, such that there exist an étale covering S' of S and a

Galois covering Q of $X' = X \times_S S'$ such that one has an isomorphism $P \simeq Q \times_{X'} \tilde{X}$. For the sequence (4.1.1) to be exact, it is necessary and sufficient that the canonical morphism

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \longrightarrow K^{\mathbb{L}}$$

be surjective. By the interpretation of $K^{\mathbb{L}}$ this amounts to saying that, for every étale covering S' of S and for every Galois covering Q of X' whose group is an $\mathbb{L}$ -group, such that $P = Q \times_{X'} \tilde{X}$ is connected, $Q|_{X_{\bar{s}}}$ is then connected. Let us show that this last condition is satisfied. Indeed, let S' be an étale covering of S , Q a Galois covering of X' whose group is an $\mathbb{L}$ -group F , such that $Q|_{X_{\bar{s}}}$ is disconnected, and let us show that, after replacing S' by an étale covering, Q becomes disconnected. There exist a subgroup G of F distinct from F and a torsor R under $G|_{X_{\bar{s}}}$ such that $Q|_{X_{\bar{s}}}$ is obtained by extension of the structure group $G \rightarrow F$ from R . The étale covering $E = Q/G$ of X' is such that $E|_{X_{\bar{s}}}$ has a section. By 1.16 f_*E is locally constant constructible and, after replacing S' by an étale covering, one may even assume that f_*E is constant. Since (E, f') is cohomologically proper relative to S' in dimension ≤ 0 and since $H^0(X_{\bar{s}}, E|_{X_{\bar{s}}})$ is nonempty, we see that E has a section. But this proves that Q is disconnected, which completes the proof.

From 4.1 we deduce the following lemma, which will be used in 4.6.

LEMMA 4.2. *Let S be a connected scheme, $f: X \rightarrow S$ a locally 0-acyclic and 0-acyclic morphism, $\mathbb{L}$ a set of prime numbers. Suppose that, for every finite constant sheaf of $\mathbb{L}$ -groups F on X , (F, f) is cohomologically proper relative to S in dimension ≤ 1 , and that, for every étale covering S' of S and for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$ -group, f'_*E is constructible. Then, if $\bar{s}$ is a geometric point of S and a a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \longrightarrow \pi'_1(X, a) \longrightarrow \pi_1(S, a) \longrightarrow 1$$

is exact.

the hypotheses of 4.1 are satisfied. It follows indeed from 1.13 3) that, for every scheme S' étale over S and for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$ -group, (E, f') is cohomologically proper relative to S' in dimension ≤ 0 .

PROPOSITION 4.3. *Let S be a connected scheme, $\mathbb{L}$ a set of prime numbers, $f: X \rightarrow S$ a 0-acyclic morphism, locally 1-aspherical for $\mathbb{L}$ (SGA 4 XV 1.11), $g: S \rightarrow X$ a section of f . Let $\bar{s}$ be a geometric point of S , a a geometric point of the fiber $X_{\bar{s}}$. Suppose that, for every constant sheaf of $\mathbb{L}$ -groups F , (F, f) is cohomologically proper in dimension ≤ 1 , that the direct image under f of the stack of torsors under F is a 1-constructible stack (0) and that, for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$ -group, f'_*E is constructible. Then the sequence of group homomorphisms*

$$(4.3.1) \quad 1 \longrightarrow \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a) \longrightarrow 1$$

is exact.

Taking 4.2 into account, it suffices to show the injectivity of the morphism u , i.e. to prove that, for every principal covering $\bar{Z}$ of $X_{\bar{s}}$ whose group is an $\mathbb{L}$ -group C , there exist an étale covering Z of X and a morphism from a connected component of $Z|_{X_{\bar{s}}}$ to $\bar{Z}$ (V 6.8). So let $\bar{Z}$ be a principal covering of $X_{\bar{s}}$ whose group is an $\mathbb{L}$ -group C , and $\bar{z}$ its class in $H^1(X_{\bar{s}}, C_{X_{\bar{s}}})$. By 1.5 d), we have a canonical isomorphism

$$\left(R^1 f_* C_X \right)_{\bar{s}} \xrightarrow{\sim} H^1(X_{\bar{s}}, C_{X_{\bar{s}}}),$$

and by 1.16, $R^1 f_* C_X$ is a locally constant constructible sheaf. One can therefore find an étale covering S' of S such that $R^1 f_* C_X|_{S'}$ is constant. If $\bar{s} \rightarrow S'$ is a geometric point above the geometric point $\bar{s} \rightarrow S$, there exists an element z of $H^0(S', R^1 f_* C_X)$ whose image in $H^1(X_{\bar{s}}, C_{X_{\bar{s}}})$ is $\bar{z}$. By Lemma 4.3.1 below, one can find an étale covering S'_1 of S' and a torsor P on X'_1 with group C whose image in $H^0(S'_1, R^1 f_* C)$ is equal to the restriction of z .

The torsor P is representable by an étale covering Z of $X'_1 = X \times_S S'_1$ such that $Z \times_{X'_1} X_{\bar{s}}$ is isomorphic to $\bar{Z}$. If one regards Z as an étale covering of X , one then has a morphism from $Z \times_X X_{\bar{s}}$ to $\bar{Z}$, which completes the proof.

LEMMA 4.3.1. *Let $f: X \rightarrow S$ be a 0-acyclic and locally 0-acyclic morphism, g a section of f . Let C be a finite constant group such that (C_X, f) is cohomologically proper in dimension ≤ 0 and such that the direct image under f of the stack of torsors under C_X is constructible. Then, for every section z of $H^0(S, R^1 f_* C_X)$, one can find an étale covering S_1 of S and, if $X_1 = X \times_S S_1$, an element of $H^1(X_1, C_{X_1})$ whose image under the canonical morphism*

$$H^1(X_1, C_{X_1}) \longrightarrow H^0(S_1, R^1 f_* C_{X_1})$$

is equal to the restriction of z to $H^0(S_1, R^1 f_ C_{X_1})$.*

For every scheme S' étale over S , we set $X' = X \times_S S'$, and we denote by g' (resp. F' , etc.) the inverse image of g (resp. F , etc.) under the morphism $S' \rightarrow S$. The presheaf G on S defined by

$$G(S') = \left\{ \begin{array}{l} \text{classes mod. isomorphism of torsors } P \text{ on } X' \text{ with group } C_{X'}, \text{ endowed with} \\ \text{an isomorphism } g'^* P \xrightarrow{\sim} C_{S'} \end{array} \right\}$$

is then a sheaf. Indeed, this follows by descent from the fact that an isomorphism of a torsor P on X' is completely determined by its restriction to $g'(S')$. Moreover, we have a surjective morphism

$$G \longrightarrow R^1 f_* F.$$

Let z be an element of $H^0(S, R^1 f_* C_X)$ and H the subsheaf of G that is the inverse image of z . It suffices to show that H is a locally constant constructible sheaf. Now, this property being local on S , we may assume that z comes from an element of $H^1(X, C_X)$ represented by a torsor P such that $g^* P$ is isomorphic to C_S . Giving an isomorphism $i: g^* P \xrightarrow{\sim} C_S$ amounts to giving a global section of $\mathbf{Aut}_{C_S}(g^* P)$, and two isomorphisms i and i' define the same element of $G(X)$ if and only if ii'^{-1} is the image of an element of $\mathbf{Aut}_{C_X}(P)$. If one considers the canonical injection

$$f_* \mathbf{Aut}_{C_X}(P) \longrightarrow \mathbf{Aut}_{C_S}(g^* P) \simeq C,$$

H is therefore identified with the quotient of $\mathbf{Aut}_{C_S}(g^* P)$ by $f_* \mathbf{Aut}_{C_X}(P)$. By 1.16 $f_* \mathbf{Aut}_{C_X}(P)$ is locally constant; hence the same holds for H , which completes the proof.

EXAMPLES 4.4. Let us note that, if S is a connected scheme, the hypotheses of 4.1 are satisfied when the morphism f is proper, flat, of finite presentation, with separable connected geometric fibers, $\mathbb{L}$ being arbitrary (cf. X 1.3). The hypotheses of 4.3 are satisfied if moreover f is smooth and has a section, if one denotes by $\mathbb{L}$ the set of prime numbers distinct from the residue characteristics of S (SGA 4 XV 2.1 and XVI 5.2).

The hypotheses of 4.1 are also satisfied if S is connected, if one has a scheme Z proper of finite presentation, flat over S , with separable connected geometric fibers, such that X is the complement in Z of a divisor with normal crossings relative to S , $\mathbb{L}$ being the set of prime numbers distinct from the residue characteristics of S (2.9). The hypotheses of 4.3 are satisfied if moreover f is smooth and has a section.

4.5. Let us take up again the notation and hypotheses of 4.3. If $\bar{s}$ is a geometric point of S and $a = g(\bar{s})$, the section g allows one to define a morphism

$$w: \pi_1(S, a) \longrightarrow \pi'_1(X, a),$$

so that $\pi'_1(X, a)$ is identified with the semi-direct product of $\pi_1(S, a)$ by $\pi_1(X_{\bar{s}}, a)$. The profinite group $\pi_1(S, a)$ therefore acts on $\pi_1(X_{\bar{s}}, a)$. Since $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is a strict projective limit of groups invariant under the action of $\pi_1(S, a)$, the datum of $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ endowed with this action is equivalent to the datum of a strict projective system of finite étale group schemes over S , which we denote by

$$\pi_1^{\mathbb{L}}(X/S, g, \bar{s}) \quad \text{or simply} \quad \pi_1^{\mathbb{L}}(X/S, g).$$

We then have the following properties:

4.5.1. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$ -groups, the set E of classes of torsors P under the inverse image G_X of G on X , endowed with an isomorphism $g^*P \xrightarrow{\sim} G$, is canonically isomorphic to the set

$$\mathrm{Hom}_S(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \quad \text{mod. inner automorphisms of } G.$$

4.5.2. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$ -groups, the sheaf $R^1f_*G_X$ is canonically isomorphic to the sheaf associated with the presheaf

$$S' \longmapsto \mathrm{Hom}_{S'}(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \quad \text{mod. inner automorphisms of } G.$$

(S' denotes a scheme étale over S).

4.5.3. Let S' be a connected S -scheme, $\bar{s}$ a geometric point of S' , X' , g' the respective inverse images of X , g over S' . Then $\pi_1^{\mathbb{L}}(X'/S', g', \bar{s})$ is canonically isomorphic to the inverse image of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$ over S' . For every geometric point ξ of S , the fiber $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})_{\xi}$ is isomorphic to $\pi_1^{\mathbb{L}}(X_{\xi})$.

Indeed, the datum of G is equivalent to the datum of an abstract $\mathbb{L}$ -group $\mathbf{G}$ on which $\pi_1(S, a)$ acts, whence an action of $\pi_1^{\mathbb{L}}(X, a)$ on $\mathbf{G}$. The isomorphism defined in 4.5.1 is then obtained by restriction to the subset E from the canonical morphism

$$H^1(\pi_1^{\mathbb{L}}(X, a), \mathbf{G}) \longrightarrow H^1(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G}) = \mathrm{Hom}(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G}) / \text{inn. aut. } \mathbf{G},$$

the set E being mapped bijectively onto the subset of morphisms from $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ to $\mathbf{G}$ that are compatible with the action of $\pi_1(S, a)$. Assertion 4.5.3 then follows from the definition of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$ taking into account the homotopy exact sequence (4.3.1), and 4.5.2 is deduced from 4.5.1 and 4.5.3.

PROPOSITION 4.6 (Künneth formula). *Let k be a separably closed field of characteristic $p \geq 0$, X and Y two connected k -schemes, a a geometric point of X , b a geometric point of Y , c a geometric point of $X \times_k Y$ above a and b . Suppose that one of the following two conditions is satisfied:*

- a) *X is of finite type over k and the schemes of finite type over an algebraic closure $\bar{k}$ of k , of dimension $\leq \dim X$, are strongly desingularizable. (SGA 5 I 3.1.5).*
- b) *X is quasi-compact and quasi-separated and every scheme of finite type over $\bar{k}$ is strongly desingularizable.*

Then, if p' is the set of prime numbers distinct from p , the morphism

$$(4.6.0) \quad \pi_1^{p'}(X \times_k Y, c) \longrightarrow \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b),$$

deduced from the homomorphisms on the fundamental groups associated with the projections

$$X \times_k Y \longrightarrow X \quad \text{and} \quad X \times_k Y \longrightarrow Y,$$

is an isomorphism.

We may assume k algebraically closed and X reduced (SGA 4 VIII 1.1). Let $Z = X \times_k Y$, and let $g: X \rightarrow k$ and $f: Z \rightarrow Y$ be the canonical morphisms. The morphism g , hence also f , is universally locally 1-aspherical for p' by 3.5. Since X is connected, f is 0-acyclic (SGA 4 XV 1.16). On the other hand it follows from 3.2 that, for every finite p' -group C , (C_X, g) is cohomologically proper relative to k in dimension ≤ 1 . It follows that (C_Z, f) is cohomologically proper relative to Y in dimension ≤ 1 (1.5 c)) and that f_*C_Z and $R^1f_*C_Z$ are constant sheaves. Consequently g satisfies all the hypotheses of 4.2. We therefore have the exact sequence

$$\pi_1^{p'}(X_b, c) \longrightarrow \pi_1^{p'}(Z, c) \longrightarrow \pi_1^{p'}(Y, b) \longrightarrow 1.$$

Moreover the composite morphism

$$\pi_1^{p'}(X_b, c) \longrightarrow \pi_1^{p'}(Z, c) \longrightarrow \pi_1^{p'}(Y, b)$$

is an isomorphism and we therefore have the exact sequence

$$1 \longrightarrow \pi_1^{p'}(X, a) \longrightarrow \pi_1^{p'}(Z, c) \longrightarrow \pi_1^{p'}(Y, b) \longrightarrow 1.$$

On the other hand the morphism (4.6.0) allows one to define a morphism from this exact sequence to the exact sequence

$$1 \longrightarrow \pi_1^{p'}(X, a) \longrightarrow \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b) \times \pi_1^{p'}(Y, b) \longrightarrow 1,$$

and it follows that the morphism (4.6.0) is an isomorphism.

4.7. Let

$$\begin{array}{ccccc} X & \longleftarrow & X_U & \longleftarrow & X_{\bar{s}} \\ f \downarrow & & f_U \downarrow & & \downarrow \\ S & \longleftarrow & U & \longleftarrow & \bar{s} \end{array}$$

be a diagram whose squares are cartesian, where S is an arcwise connected scheme (SGA 4 IX 2.12), U a connected open subset of S , $\bar{s}$ a geometric point of U . Let a be a geometric point of $X_{\bar{s}}$, $\mathbb{L}$ a set of prime numbers. Let g be a section of f and suppose that the following conditions are satisfied:

- a) The morphism f is 0-acyclic, locally 0-acyclic, and, for every étale covering S' of S and every étale covering E of $X \times_S S'$ that is a quotient of a Galois covering whose group is an $\mathbb{L}$ -group, $(E, f_{(S')})$ is cohomologically proper relative to S' in dimension ≤ 0 .
- b) The morphism f_U is locally 1-aspherical for $\mathbb{L}$, and, for every finite constant sheaf of $\mathbb{L}$ -groups F on X_U , (F, f_U) is cohomologically proper in dimension ≤ 1 and the fibers of $R^1 f_{U*} F$ are finite.

From 4.1 and 4.3 one then deduces the following commutative diagram, whose rows are exact:

$$(4.7.0) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) & \longrightarrow & \pi_1'(X_U, a) & \longrightarrow & \pi_1(U, a) \longrightarrow 1 \\ & & \parallel & & \downarrow & & \downarrow \\ & & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) & \longrightarrow & \pi_1'(X, a) & \longrightarrow & \pi_1(S, a) \longrightarrow 1 \end{array}$$

Thanks to the section g , we have morphisms

$$\pi_1(U, a) \longrightarrow \pi_1'(X_U, a), \quad \pi_1(S, a) \longrightarrow \pi_1'(X, a)$$

from which one deduces a morphism from the amalgamated sum of $\pi_1(S, a)$ and $\pi_1'(X_U, a)$ over $\pi_1(U, a)$ to $\pi_1'(X, a)$:

$$(4.7.1) \quad \varphi: \pi = \pi_1(S, a) \coprod_{\pi_1(U, a)} \pi_1'(X_U, a) \longrightarrow \pi_1'(X, a).$$

Suppose that the following condition is satisfied:

- c) If $T = S - U$, we have $\text{prof ét}_T(S) \geq 2$ (SGA 2 XIV 1.1).

Then the functor which associates with an étale covering of S its restriction to U is fully faithful (SGA 2 XVI 1.4). It follows that the morphism $\pi_1(U, a) \rightarrow \pi_1(S, a)$ is surjective (V 6.9) and one deduces from the diagram (4.7.0) that the same holds for the morphism $\pi_1'(X_U, a) \rightarrow \pi_1'(X, a)$; a fortiori φ is an epimorphism. Let

$$(4.7.2) \quad K = \text{Ker}(\pi_1(U, a) \longrightarrow \pi_1(S, a)).$$

The group π of (4.7.1) is identified with the quotient of $\pi_1'(X_U, a)$ by the closed invariant subgroup generated by the image L of K in $\pi_1'(X_U, a)$. Let us regard $\pi_1'(X_U, a)$ as the semi-direct product of $\pi_1(U, a)$ by $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$. The group K then acts by inner automorphisms on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$, and the quotient $\pi = \pi_1'(X_U, a)/L$ is identified with the semi-direct product of

$\pi_1(U, a)/K = \pi_1(S, a)$ by the group $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ of coinvariants of $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ under K . We finally have an epimorphism

$$(4.7.3) \quad \varphi: \pi = \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K \cdot \pi_1(S, a) \longrightarrow \pi'_1(X, a).$$

The proposition which follows gives conditions under which the morphism φ is an isomorphism.

PROPOSITION 4.7.4. *The notation is that of 4.7. We suppose that, in addition to conditions a), b), c), the following conditions are satisfied:*

- d) *For every point t of $T = S - U$, the morphism f is locally 1-aspherical for $\mathbb{L}$ at $g(t)$.*
- e) *For every point $t \in T$, every irreducible component of the fiber X_t contains $g(t)$ and, for every point x of $X_t - \{g(t)\}$ which is not maximal, we have*

$$\text{prof } \text{hop}_x(X) \geq 3 \quad (\text{SGA 2 XIV 1.2})$$

and the ring $\mathcal{O}_{X,x}$ is noetherian.

Then the morphism (4.7.3) is an isomorphism.

As was said previously, the group π is identified with the quotient of $\pi'_1(X_U, a)$ by the closed invariant subgroup L generated by the image of K (4.7.2) in $\pi'_1(X_U, a)$. This amounts to saying that π classifies the principal coverings Z of X_U such that $g_U^{-1}(Z)$ extends to an étale covering of S , and which induce on $X_{\bar{s}}$ a covering obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$ -group. To prove that φ is an isomorphism, it suffices to show that such a covering Z extends to all of X .

Let us first show that Z extends to an étale covering of an open set containing X_U and $g(S)$. Let W be a scheme étale over X whose image contains X_U and $g(S)$, and let us set $W_U = W \times_S U$. From the fact that the morphism $W \rightarrow S$ is 0-acyclic and from the relation $\text{prof. ét}_T S \geq 2$, it follows that we have

$$\text{prof. ét}_{(W-W_U)} W \geq 2$$

(SGA 2 XIV 1.13); consequently, if $Z|_{W_U}$ extends to an étale covering of W , this extension is unique up to unique isomorphism. It follows that the problem of extending Z to a neighborhood of $g(S) \cup X_U$ is local for the étale topology in the neighborhood of the points of $g(T)$. If t is a point of T , we set $x = g(t)$, and we denote by $\bar{X}$ (resp. $\bar{S}$) the strict localization of X at $\bar{x}$ (resp. of S at $\bar{t}$), $\bar{U} = U \times_S \bar{S}$, $\bar{X}_U = \bar{X} \times_X X_U$, $\bar{g}: \bar{S} \rightarrow \bar{X}$ the morphism deduced from g . It suffices to show that, for every point t of T , the inverse image $\bar{Z}$ of Z on $\bar{X}_U$ extends to $\bar{X}$ or, what amounts to the same, is trivial. Now, by definition of Z , the inverse image of $\bar{Z}$ on $\bar{U}$ is trivial. To prove that $\bar{Z}$ is trivial, it suffices to show that it is of the form $\bar{f}_U^* E$, where E is a principal covering of $\bar{U}$; indeed we shall then have $E \simeq \bar{g}_U^* \bar{f}_U^* E \simeq \bar{g}_U^* \bar{Z}$, whence the result since $\bar{g}_U^* \bar{Z}$ is trivial. Now, the morphism $\bar{f}_U$ being 0-acyclic and locally 0-acyclic, it suffices, in order to prove that $\bar{Z}$ comes from $\bar{U}$, to show that, for every geometric point algebraic over a point of $\bar{U}$, which one may assume to be the point $\bar{s}$, $\bar{Z}|_{X_{\bar{s}}}$ is trivial (SGA 4 XV 1.15). But since $\bar{Z}|_{X_{\bar{s}}}$ is obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$ -group, this follows from the fact that the morphism $\bar{f}$ is 1-aspherical for $\mathbb{L}$.

We have therefore shown that there exists an open neighborhood V of $g(S) \cup X_U$ such that Z extends to an étale covering Z_V of V . Let us show that Z_V extends to all of X . It suffices to see that, for every point x of $X - V$, we have

$$\text{prof } \text{hop}_x X \geq 3.$$

Now this follows from hypothesis e) and from the fact that a point x of $X - V$ cannot be maximal in its fiber X_t since, every irreducible component of X_t containing $g(t)$, every maximal point of X_t belongs to V .

COROLLARY 4.8. *The hypotheses are those of 4.7.4, but we assume moreover that $\pi_1(S, a) = 1$. Then we have an isomorphism*

$$(*) \quad \pi_1^{\mathbb{L}}(X, a) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K.$$

In particular, if $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is topologically of finite presentation and if K acts on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ through a group of finite type, then $\pi_1^{\mathbb{L}}(X, a)$ is topologically of finite presentation.

The isomorphism $(*)$ was proved in 4.7.4. Suppose that $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is the quotient of the free pro- $\mathbb{L}$ -group on n generators $L(x_1, \dots, x_n)$ by the closed invariant subgroup generated by the elements $y_1, \dots, y_p$ of $L(x_1, \dots, x_n)$, and suppose that K acts through a group generated by elements $k_1, \dots, k_q$. If for every $i \in [1, n]$, $j \in [1, q]$, we denote by z_{ij} an element of $L(x_1, \dots, x_n)$ lifting the element $(k_j \cdot x_i)x_i^{-1}$, then $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ is the quotient of $L(x_1, \dots, x_n)$ by the closed invariant subgroup generated by the elements $(y_i)_{i \in [1, p]}, (z_{ij})_{i \in [1, n], j \in [1, q]}$.

REMARKS 4.9. a) Conditions a) to e) of 4.7 are satisfied when S is a connected normal scheme, U a dense retrocompact open subset of S , f a proper morphism of finite presentation, with fibers that are geometrically connected and irreducible at every point t of T , f being moreover separable, smooth at the points of $X_U \cup g(T)$, $\mathbb{L}$ being the set of prime numbers distinct from the residue characteristics of S , and X being regular at every point of X_t . Condition a) follows indeed from SGA 4 XV 4.1 and 1.4; conditions b) and d) follow from 1.4 and SGA 4 XV 2.1 and XVI 5.2. Finally e) follows from SGA XIV 1.11.

b) Corollary 4.5 applies to compute the fundamental group $\pi_1^{p'}(X)$ of a proper and smooth surface X over a separably closed field k of characteristic p (p' denoting the set of prime numbers distinct from p). The method was communicated to us by J.P. MURRE; it consists in reducing, by blowing up X , to the case where one has a fibration $X \rightarrow \mathbf{P}_k^1$ and an open set U of $\mathbf{P}_k^1$ satisfying the hypotheses of 4.7 (see SGA 7 for more details). The same method can be used more generally (loc. cit.) to prove that, if X is a connected k -scheme of finite type, and if the schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k are strongly desingularizable (SGA 5 I 3.1.5), then $\pi_1^{p'}(X)$ is topologically of finite presentation.

5. Appendix I: Variations on Abhyankar's lemma

This appendix contains various variants of Abhyankar's lemma.

PROPOSITION 5.1. *Let $X = \text{Spec } A$ be a regular local scheme, $D = \sum_{1 \leq i \leq r} \text{div } f_i$ a divisor with normal crossings, where the f_i are elements of the maximal ideal of A forming part of a regular system of parameters. Let n_i , $1 \leq i \leq r$, be integers ≥ 0 and set*

$$X' = X[T_1, \dots, T_r] / (T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

$U' = U \times_X X'$. Then X' is regular and U' is the complement in X' of the divisor with normal crossings $\sum_{1 \leq i \leq r} \text{div } T_i$. If the integers n_i are prime to the residue characteristic p of X , U' is a connected étale covering of U , tamely ramified relative to D (2.3 c)).

Indeed X' is the spectrum of a local ring A' whose maximal ideal is generated by $T_1, \dots, T_r$. (We may assume $r = \dim(A)$.) Since A' is finite and flat over A , hence of dimension r , A' is regular (EGA 0_{IV} 17.1.1) and the T_i form a regular system of parameters of A' . Suppose the n_i prime to p . Since all the f_i are invertible on U , the fact that U' is étale over U follows from I 7.4. Moreover U' is tamely ramified relative to D ; indeed, let x_i be the generic point of $V(f_i)$, $\bar{R}$ the strict localization of $R = \mathcal{O}_{X, x_i}$, $\bar{K}$ the field of fractions of $\bar{R}$. Then the $\bar{K}$ -algebra that represents $U'|\bar{K}$ is obtained from the field $\bar{K}[T_i]/(T_i^{n_i} - f_i)$ by making an unramified extension; it is therefore tamely ramified over $\bar{R}$.

PROPOSITION 5.2 (Absolute Abhyankar lemma). *Let X be a regular local scheme,*

$$D = \sum_{1 \leq i \leq r} \text{div } f_i$$

a divisor with normal crossings as in 5.1, $Y = \text{Supp } D$, $U = X - Y$. Let V be an étale covering of U tamely ramified relative to D . If x_i is the generic point of the closed set $V(f_i)$, $\mathcal{O}_{X, x_i}$ is a discrete valuation ring with field of fractions K_i , and we have $V|_{K_i} = \text{Spec}(\prod_{j \in J_i} L_j)$, where the L_j are finite separable extensions of K_i ; denote by n_j the order of the inertia group of a Galois extension generated by L_j and by n_i the l.c.m. of the n_j as j runs through J_i . If we set

$$X' = X[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

$U' = U_{(X')}$, $V' = V_{(X')}$, etc., the étale covering V' of U' extends in a unique way, up to unique isomorphism, to an étale covering of X' , and the n_i are prime to the residue characteristic p of X .

Uniqueness follows from the fact that X' is normal (5.1); indeed an étale covering of X' that extends V' is isomorphic to the normalization of X' in the fiber of V' at the generic point of X' (10.2). If $\bar{x}'$ is a geometric point of Y' , we denote by $\bar{X}'$ the strict localization of X' at $\bar{x}'$, $\bar{V}' = V'_{(\bar{X}')}$, etc. By descent, taking uniqueness into account, it suffices to show

that, for every geometric point $\bar{x}'$ of Y' , the étale covering $\bar{V}'$ of $\bar{U}'$ extends to $\bar{X}'$. Given that an étale covering of an open subset of the regular scheme X' which contains all the points x' such that $\dim \mathcal{O}_{X', x'} \leq 1$ extends to all of X (SGA 2 XIV 1.11), we may even restrict ourselves to the points $\bar{x}'$ which project onto a maximal point of Y' . Now, at such a point $\bar{x}'$, the fact that $\bar{V}'$ extends to an étale covering of $\bar{X}'$ follows from 3.6.

Let us show that the n_i are prime to p . Indeed, if this were not so, we would have for example $p|n_1$. Replacing X by

$$X[T_1, \dots, T_r]/(T_1^{n_1/p} - f_1, T_2^{n_2} - f_2, \dots, T_r^{n_r} - f_r),$$

we reduce to the case where $X' = X[T_1]/(T_1^p - f_1)$. It suffices to show that V extends to an étale covering of X , because we will then have $n_1 = 1$, contrary to the hypothesis. For this we may assume that X is strictly local. Let Z be the closed subscheme of X with equation $p = 0$ and $Z_1 = Z \cap X_{f_1}$; Z_1 is a nonempty open subset of Z . By what precedes, the étale covering V' of U' extends to an étale covering W' of X' . Let W_1'' and W_2'' be the inverse images of W' under the two projections $X'' = X' \times_X X' \rightrightarrows X'$, and let us show that the descent isomorphism $u: W_1''|_{U''} \rightarrow W_2''|_{U''}$ extends to an X'' -morphism $W_1'' \rightarrow W_2''$, which will necessarily be a descent datum on W' relative to $X' \rightarrow X$. Let Z'' (resp. Z_1'') be the inverse image of Z (resp. Z_1) in X'' . Since the morphism $Z'' \rightarrow Z$ is radicial, there exists an isomorphism $v: W_1''|_{Z''} \rightarrow W_2''|_{Z''}$ which extends the isomorphism $u|_{Z_1''}$. But, since X is henselian, we have a bijection

$$\text{Hom}_{X''}(W_1'', W_2'') \simeq \text{Hom}_{Z''}(W_1''|_{Z''}, W_2''|_{Z''}),$$

whence a morphism $w: W_1'' \rightarrow W_2''$ lifting v . The subscheme, at once open and closed, of X'' over which u and w coincide contains Z_1'' , hence is equal to X'' , whence the fact that V extends to X .

5.3.0. Let us take up again the hypotheses and the notation of 5.2, assuming moreover X strictly local. Then it follows from loc. cit. that every connected étale covering of U tamely ramified relative to D is a quotient of a tamely ramified covering of the form

$$U' = U[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

where the n_i are integers prime to p . Let μ_n be the group of n th roots of unity of U . The group of U -automorphisms of U' is none other than the group $\mu_{n_1} \times \dots \times \mu_{n_r}$, an n_i th root of unity ξ_i acting on U' by transforming T_i into $\xi_i T_i$. We therefore have the following statement:

COROLLARY 5.3. *Let X be a regular strictly local scheme of residue characteristic $p \geq 0$, $D = \sum_{1 \leq i \leq n} \text{div } f_i$ a divisor with normal crossings on X , $U = X - \text{Supp } D$. Set*

$$\tilde{U} = \varprojlim_{(n_i)} U[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

the projective limit being taken over the filtered ordered set (for the divisibility relation) of families of integers $n_i > 0$, prime to p . Then $\tilde{U}$ is a tamely ramified universal covering of U . Consequently the tamely ramified fundamental group of U is

$$\pi_1^t(U) \simeq \prod_{\ell \neq p} \mathbf{Z}_\ell[1]^r \quad (\text{canonical isomorphism})$$

where we have set $\mathbf{Z}_\ell[1] = \varprojlim_{n>0} \mu_{\ell^n}$. The group $\pi_1^t(U)$ is non-canonically isomorphic to $\prod_{\ell \neq p} \mathbf{Z}_\ell^r$.

PROPOSITION 5.4. *Let $f: X \rightarrow S$ be a morphism of schemes, $D = \sum_{1 \leq i \leq r} \text{div } f_i$ a divisor with normal crossings relative to S (2.1), where, for each point x of $Y = \text{Supp } D$, if $I(x) \subset [1, r]$ is the set of i such that $f_i(x) = 0$, the subscheme $V((f_i)_{i \in I(x)})$ is smooth over S of codimension $\text{card } I(x)$ in X . Let $U = X - Y$. Let x be a point of Y , $X_1 = \text{Spec } \mathcal{O}_{X,x}$, $U_1 = U \times_X X_1$, n_i , $i \in I(x)$, integers, and*

$$X' = X[T_i]_{i \in I(x)} / (T_i^{n_i} - f_i).$$

Then, if x' is the point of X' over x , X' is smooth over S at x' . If the integers n_i are prime to the characteristic p of $k(x)$, $U'_1 = U_1 \times_X X'$ is a connected étale covering of U_1 tamely ramified over X_1 relative to S_1 (2.1.1).

If $s = f(x)$, the geometric fiber X'_s is regular at the point x' (5.1); since X' is flat over S in a neighborhood of Y , this proves that X' is smooth over S at x' (EGA IV 12.1.6). If the integers n_i are prime to p , U'_1 is an étale covering of U_1 (7.4); it is tamely ramified over X_1 relative to S because this is so on the geometric fibers at each point of S (5.1). Finally the fact that U'_1 is connected follows from (SGA 4 XVI 3.2).

PROPOSITION 5.5 (Relative Abhyankar lemma). *Let X be an S -scheme, D a divisor with normal crossings relative to S , as in (5.4). Let $Y = X - \text{Supp } D$, $U = X - Y$, x a point of Y , X_1 the strict localization of X at a geometric point over x , $U_1 = U \times_X X_1$, V_1 an étale covering of U_1 . Assume that, for every maximal point s of S , $V_{1\bar{s}}$ is tamely ramified over $X_{1\bar{s}}$ relative to $\bar{s}$. Then one can find integers n_i prime to the characteristic p of $k(x)$, with $i \in I(x)$, such that, if we set*

$$X'_1 = X_1[T_i]_{i \in I(x)} / (T_i^{n_i} - f_i),$$

$U'_1 = U_1 \times_{X_1} X'_1$, etc., the étale covering V'_1 of U'_1 extends in a unique way, up to unique isomorphism, to an étale covering of X'_1 . In particular V_1 is tamely ramified over X_1 relative to S .

We may assume S local noetherian with closed point $f(x)$. For each maximal point s of S and for each $i \in I(x)$, let x_i be the generic point of the closed set $V(f_i)$ of the fiber $X_{1\bar{s}}$. The local ring $(\mathcal{O}_{X_1, x_i})_{\text{red}}$ is a discrete valuation ring with field of fractions K_i and we have $V_1|K_i = \text{Spec}(\prod_{j \in I(x_i)} L_j)$, where L_j is a finite separable extension of K ; we denote by n_j the order of the inertia group of a Galois extension generated by L_j and by n_i the l.c.m. of the n_j as s runs through the maximal points of S and $j \in I(x_i)$.

The n_i being thus chosen, we shall show that V'_1 extends in a unique way to an étale covering of X'_1 . Uniqueness follows from the fact that, X' being smooth over S at the points of Y , we have $\text{prof } \text{ét}_{Y'_1}(X'_1) \geq 2$ (SGA 4 XVI 3.2 or SGA 2 XIV 1.19). Let x'_1 be a point of Y'_1 , $\bar{x}'_1$ a geometric point over x'_1 , and denote by $\overline{X'_1}$ the strict localization of X'_1 at $\bar{x}'_1$, $\overline{U'_1} = U'_{1(\overline{X'_1})}$, etc. By descent, taking uniqueness into account, it suffices to show that $\overline{V'_1}$ extends to $\overline{X'_1}$. Moreover we may restrict ourselves to taking for x'_1 the maximal points of Y'_1 ; indeed we will then have an extension of V'_1 over an open subset W'_1 of X'_1 containing the maximal points of Y'_1 ; now, if $Z'_1 = X'_1 - W'_1$, we have $\text{codim}(Z'_{1s}, X'_{1s}) \geq 2$ if s is a maximal point of S and $\text{codim}(Z'_{1s}, X'_{1s}) \geq 1$, $\text{prof } \text{ét}_s(S) \geq 1$ if s is a point of S which is not maximal; the fact that V'_1 extends to all of X'_1 then follows from (SGA 2 XIV 1.20). But, at a geometric

point $\bar{x}'_1$ over a maximal point of Y'_1 , $X'_{1\text{red}}$ is the spectrum of a discrete valuation ring, and the fact that $\bar{V}'_1$ extends to $\bar{X}'_1$ then follows from X 3.6.

Let us show that the n_i are prime to p . Indeed, if this were not so, there would be an index $i_0 \in I(x)$ such that p divides n_{i_0} . Replacing X by $X_1[T_{i_0}, T_i]_{i \in I(x)} / (T_i^{n_{i_0}/p} - f_{i_0}, T_i^{n_i} - f_i)$, we reduce to the case where $X'_1 = X_1[T]/T^p - f_{i_0}$. By what precedes, the étale covering V'_1 of U'_1 extends to an étale covering E'_1 of X'_1 . Let η be the closed point of S ; since the morphism $X'_{1\eta} \rightarrow X_{1\eta}$ is radical, $V_{1\eta}$ extends to an étale covering $E_{1\eta}$ of $X_{1\eta}$. One then deduces, as in 5.2, that E'_1 is endowed with a descent datum relative to the morphism $X'_1 \rightarrow X_1$, which extends the natural descent datum that we have on $E'_1|U'_1$. It follows that V_1 extends to X_1 ; but this implies that we have $n_{i_0} = 1$, contrary to the hypothesis $n_{i_0} = p$.

COROLLARY 5.6. *Let X be an S -scheme, $D = \sum_{1 \leq i \leq r} \text{div } f_i$ a divisor with normal crossings relative to S , as in 5.4. Let $\bar{x}$ be a geometric point of X , $\bar{X}$ the strict localization of X at $\bar{x}$, $\bar{Y} = Y_{(\bar{X})}$, $\bar{U} = \bar{X} - \bar{Y}$ and*

$$\tilde{U} = \varprojlim_{(n_i)} \bar{U}[T_i]_{i \in I(x)} / (T_i^{n_i} - f_i),$$

the projective limit being taken over the filtered set of families of integers $n_i > 0$, prime to the characteristic p of $k(x)$. Then $\tilde{U}$ is a tamely ramified universal covering of $\bar{U}$ relative to S . Consequently the tamely ramified fundamental group of $\bar{U}$ is

$$\pi_1^t(\bar{U}) \simeq \prod_{\ell \neq p} \mathbf{Z}_\ell[1]^{I(x)} \quad (\text{canonical isomorphism}).$$

The group $\pi_1^t(\bar{U})$ is non-canonically isomorphic to $\prod_{\ell \neq p} \mathbf{Z}_\ell^{I(x)}$.

REMARK 5.6.1. Let X be an S -scheme, $D = \sum_{1 \leq i \leq r} \text{div } f_i$ a divisor with normal crossings relative to S , as in 5.4, $U = X - \text{Supp } D$. For every subset $I \subset [1, r]$ let

$$X_I = \left(\bigcap_{i \in I} V(f_i) \right) \cap \left(\bigcap_{i \in \mathbb{C}I} X_{f_i} \right).$$

Let p be an integer that is prime or zero and let Z be a subset of X_I all of whose points are of characteristic p . Let

$$\tilde{U}_I = \varprojlim_{(n_i)} U[T_i]_{i \in I} / (T_i^{n_i} - f_i),$$

where the projective limit is taken over the filtered set of families of integers $n_i > 0$, prime to p . Then, for every geometric point $\bar{x}$ of Z , the inverse image of $\tilde{U}_I$ on $\bar{U}$ is identified with the tamely ramified universal covering of $\bar{U}$.

COROLLARY 5.7. *The notation is that of 5.6. Let $\bar{S}$ be the strict localization of S at $\bar{x}$, and*

$$\bar{g}: \bar{U} \longrightarrow \bar{S} \quad \text{and} \quad \tilde{g}: \tilde{U} \longrightarrow \tilde{S}$$

*the canonical morphisms. Then the morphisms $\bar{g}$ and $\tilde{g}$ are 0-acyclic (SGA 4 XV 1.3). Let G be a constructible sheaf of groups on $\bar{S}$, $F = \bar{g}^*G$, P a torsor under F . Then, for P to be tamely ramified over $\bar{X}$ relative to $\bar{S}$, it is necessary and sufficient that its inverse image $\tilde{P}$ on $\tilde{U}$ be trivial.*

Indeed, for every scheme $\bar{X}' = \bar{X}[T_i]_{i \in I(x)} / (T_i^{n_i} - f_i)$, where the n_i are integers > 0 prime to p , the morphism $\bar{f}': \bar{X}' \rightarrow \bar{S}$ is 0-acyclic. The geometric fibers of $\bar{f}'$ at the various points of $\bar{S}$ are therefore connected and even irreducible. The same therefore holds for the geometric fibers of the morphisms $\bar{g}': \bar{U}' \rightarrow \bar{S}$, which proves that the $\bar{g}'$, hence also $\tilde{g}$, are 0-acyclic (SGA 4 XV 1.16).

It is clear that a torsor P on $\bar{U}$ with group F , whose inverse image on $\tilde{U}$ is trivial, is tamely ramified over $\bar{X}$ relative to $\bar{S}$. Let us show that conversely, if P is tamely ramified over $\bar{X}$ relative to $\bar{S}$, its inverse image on $\tilde{U}$ is trivial.

It follows from (SGA 4 IX 2.14 (ii)) that one can find a finite morphism $n: S_1 \rightarrow \bar{S}$, a constant sheaf of groups C on S_1 , a monomorphism $G \rightarrow n_*C$. Consider the following commutative diagram made up of cartesian squares:

$$\begin{array}{ccccc} \tilde{U}_1 & \longrightarrow & U_1 & \xrightarrow{g_1} & S_1 \\ r \downarrow & & q \downarrow & & n \downarrow \\ \tilde{U} & \longrightarrow & \bar{U} & \xrightarrow{\bar{g}} & \bar{S}. \end{array}$$

Let C_1 (resp. $\tilde{C}_1$) be the inverse image of C on U_1 (resp. on $\tilde{U}_1$). We have a commutative diagram, in which i and j are isomorphisms (SGA 4 VIII 5.8):

$$(*) \quad \begin{array}{ccc} H^1(\bar{U}, q_*C_1) & \xrightarrow{i} & H^1(U_1, C_1) \\ \downarrow & & \downarrow \\ H^1(\tilde{U}, r_*\tilde{C}_1) & \xrightarrow{j} & H^1(\tilde{U}_1, \tilde{C}_1). \end{array}$$

Let Q be the torsor under q_*C_1 deduced from P by the extension of the structure group $F \rightarrow q_*C_1$. By 2.1.4, Q is tamely ramified over $\bar{X}$ relative to $\bar{S}$. To the torsor Q there corresponds, by means of i , a torsor Q_1 under C_1 , and it is clear that Q_1 is tamely ramified over $X_1 = \bar{X} \times_{\bar{S}} S_1$ relative to S_1 . It therefore follows from 5.6 that the inverse image $\tilde{Q}_1$ of Q_1 on $\tilde{U}_1$ is trivial, and diagram $(*)$ then shows that the inverse image $\tilde{Q}$ of Q on $\tilde{U}$ is trivial.

Consider the following commutative diagram, whose second row is exact (SGA 4 XII 3.1):

$$(**) \quad \begin{array}{ccccc} H^0(\bar{S}, n_*C/G) & \longrightarrow & H^1(\bar{S}, G) & = & 1 \\ k \downarrow & & \downarrow & & \\ H^0(\tilde{U}, r_*\tilde{C}_1/\tilde{F}) & \longrightarrow & H^1(\tilde{U}, \tilde{F}) & \longrightarrow & H^1(\tilde{U}, r_*\tilde{C}_1). \end{array}$$

Since the morphism $\tilde{U} \rightarrow \bar{S}$ is 0-acyclic, k is an isomorphism. The fact that $\tilde{P}$ is trivial then follows from $(**)$.

6. Appendix II: finiteness theorem for direct images of stacks

PROPOSITION 6.1. *Let S be a locally noetherian scheme, $f: X \rightarrow S$ a morphism. If S' is an S -scheme, we denote by X' (resp. f' , etc.) the inverse image of X (resp. f , etc.) under the morphism $S' \rightarrow S$. Suppose that, for every scheme S' étale over S and every constructible sheaf of sets F on X' , f'_*F is constructible, and that, for every constructible sheaf of groups F on X' , $R^1f'_*F$ is constructible. Let Φ be a 1-constructible stack on X (0). Then $f_*\Phi$ is 1-constructible.*

For every scheme S' étale over S and for every object x of $(f_*\Phi)_{S'}$, we have an isomorphism

$$\mathbf{Aut}_{S'}(x) \simeq f'_*\mathbf{Aut}_{X'}(x),$$

where, in the right-hand side of the equality, x is regarded as an object of $\Phi_{X'}$. The hypotheses made therefore imply that $f_*\Phi$ is constructible. Let $S\Phi$ be the sheaf of maximal subgerbes of Φ [1, III 2.1.7]. Since $f_*(S\Phi)$ is constructible, we can apply (SGA 4 IX 2.7) to it, and the fact that $f_*\Phi$ is 1-constructible then follows from the lemma which follows.

LEMMA 6.1.1. *Let S be a locally noetherian scheme, $f: X \rightarrow S$ a morphism, Φ a stack on X . Suppose given a sheaf on S , representable by an étale S -scheme of finite type T , a surjective morphism*

$$a: T \longrightarrow f_*(S\Phi)$$

and an object p of the fiber Φ_{X_T} (where $X_T = X \times_S T$), defining in $f_(S\Phi)(T) = S\Phi(X_T)$ an element equal to the image q under a of the identity section of $T(T)$. Let $f_T: X_T \rightarrow T$ be the*

canonical morphism and suppose that the sheaf $R^1 f_{T*}(\mathbf{Aut}_{X_T}(p))$ is constructible; then so is $S(f_*\Phi)$.

The canonical morphism $f^* f_* \Phi \rightarrow \Phi$ gives a morphism

$$S(f^* f_* \Phi) \simeq f^*(S(f_* \Phi)) \longrightarrow S\Phi,$$

whence a canonical morphism

$$\varphi: S(f_* \Phi) \longrightarrow f_*(S\Phi).$$

Let $F = S(f_* \Phi)$ and let G be the image of F under φ ; by (SGA 4 IX, 2.9) G is a constructible sheaf.

It suffices to show that, for every point s of S , there exists a nonempty open set U of s such that $F|U$ is locally constant constructible. Let $s \in S$, $\bar{s}$ a geometric point over s , $\bar{q}_1, \dots, \bar{q}_n$ the elements of $G_{\bar{s}}$. By definition of T , there exist S -morphisms $h_i: \bar{s} \rightarrow S'$ such that $\bar{q}_i = h_i^*(q)$. Let S' be the fiber product over S of n schemes isomorphic to T , $\bar{s} \rightarrow S'$ the fiber product of the h_i , $X' = X \times_S S'$, q_i (resp. p_i) the inverse image of q (resp. p) under the i th projection of S' onto T . If Ψ_i is the maximal subgerbe of $\Phi|X'$ generated by p_i , the sheaf $F_i = R^1 f'_*(\mathbf{Aut}_{X'}(p_i))$ is none other than the sheaf $S(f'_* \Psi_i)$ of maximal subgerbes of $f'_* \Psi_i$. In particular the canonical injection $\Psi_i \rightarrow \Phi|X'$ gives a morphism

$$\alpha_i: F_i \longrightarrow F|S'.$$

We shall show that α_i is a bijection of F_i onto the inverse image of q_i in $F|S'$. For every scheme S'' étale over S' , every section y of $F_i(S'')$ has as image $q_i|S''$ in $F(S'')$, because, locally for the étale topology on S'' , y is defined by an object x of $\Phi_{X''}$ which is isomorphic to $p_i|X''$. Conversely, if $y \in F(S'')$ has as image $q_i|S''$ in $F(S'')$, then locally for the étale topology on S'' , y is defined by an object x of $\Phi_{X''}$ which is isomorphic to p_i ; consequently x is an object of $\Psi_i|X''$ and consequently $y \in F_i(S'')$.

The proof is completed by using 6.1.2 below. Indeed, one can find an open neighborhood U' of s such that $q_1|U', \dots, q_n|U'$ are sections of $G(U')$ and generate this sheaf. Since the $F_i|U'$ and $G|U'$ are constructible, so is $F|U'$, by 6.1.2; replacing U by a smaller open set, $F|U$ is locally constant, which completes the proof.

LEMMA 6.1.2. *Let S be a locally noetherian scheme, $F \rightarrow G$ a surjective morphism of sheaves of groups on S . Let q_i be a finite family of sections of G over X which generate G , and, for each i , let F_i be the subsheaf of F which is the inverse image of q_i . Then, if G and the F_i are constructible, so is F .*

To prove that F is constructible, it suffices to show that, for every point s of S , there exists an open neighborhood U of s such that $F|U$ is locally constant constructible. So let s be a point of S . Since the sheaves F_i and G are constructible, one can find an open neighborhood U of s such that $F_i|U$ and $G|U$ are locally constant. Let us then show that $F|U$ is locally constant. By (SGA 4 IX 2.13 (i)), it suffices to see that, if $\bar{s}$ is a geometric point over s , $\tilde{s}$ a geometric point of U and $\bar{s} \rightarrow \tilde{s}$ a specialization morphism, the canonical morphism

$$F_{\tilde{s}} \longrightarrow F_{\bar{s}}$$

is bijective.

We consider the commutative diagrams

$$\begin{array}{ccccc} (F_i)_{\tilde{s}} & \xrightarrow{\tilde{a}_i} & F_{\tilde{s}} & \xrightarrow{\tilde{a}} & G_{\tilde{s}} \\ \downarrow \wr & & \downarrow b & & \downarrow \wr \\ (F_i)_{\bar{s}} & \xrightarrow{\bar{a}_i} & F_{\bar{s}} & \xrightarrow{\bar{a}} & G_{\bar{s}} \end{array}$$

Let $\bar{q}_i$ (resp. $\tilde{q}_i$) be the inverse image of q_i in $G_{\bar{s}}$ (resp. $G_{\tilde{s}}$); the morphisms $\bar{a}$ and $\tilde{a}$ are surjective, and the morphism $\bar{a}_i$ (resp. $\tilde{a}_i$) induces a bijection of $(F_i)_{\bar{s}}$ (resp. $(F_i)_{\tilde{s}}$) onto $\bar{a}^{-1}(q_i)$ (resp. onto $\tilde{a}^{-1}(q_i)$). It therefore follows from the diagram above that b is an isomorphism.

COROLLARY 6.2. *Let S be a locally noetherian scheme, $f: X \rightarrow S$ a proper morphism. Let Φ be a 1-constructible stack on X ; then $f_*\Phi$ is a 1-constructible stack.*

The proof of 6.1 also proves the following result, taking 2.4 2) into account.

COROLLARY 6.3. *Let S be a locally noetherian scheme, $f: X \rightarrow S$ a morphism, D a divisor on X with normal crossings relative to S (2.1), $Y = \text{Supp } D$, $U = X - Y$, $i: U \rightarrow X$ the canonical immersion. Let Φ be a stack on U given, locally for the étale topology on X and S , as the inverse image of a 1-constructible stack Ψ on S . Then the stack $i_*^t \Phi$ is 1-constructible.*

7. Bibliography

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